



SECP2523: DATABASE

SEM I 2024/2025

DATABASE DESIGN DESCRIPTIONS: LOGICAL DATABASE DESIGN FOR KADA COOPERATIVE SYSTEM

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Stakeholder: Cooperative KADA

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NOTE: A module can consist of one or multiple (related) use case. A team member in a group must be responsible to at least one module.

1.1 Overview about the company

KADA (Lembaga Kemajuan Pertanian Kemubu) is an agriculture organization in Kelantan. The workers under KADA are being given the opportunity to become cooperative members of KADA and make a loan under KADA too. But, the current system, or the as-is system in KADA to provide such services is being done manually and there is where we saw that it has very low efficiency and productivity. According to the list below, are the services that KADA provided to their staff currently.

1. Registration to become a membership of cooperative of KADA

In the current system, all the registration procedures have to undergo a manual process, such as filling in the paper form. For example, users must pick up the form in person at the headquarters in order to fill it in. In the registration process, most of the personal information of the users that are private and sensitive will be collected. Then, the users have to submit it manually also by submitting it at the headquarters. After the filled in and submitting process is done, then the registration paper would be further reviewed by the staff itself manually whether the users are qualified to become their member or not.

2. Applying a loan in cooperative of the KADA

Same as the registration process to become a membership of KADA, applying for a loan is also by filling in the paper form manually. The difference is in applying for a loan form, it would ask more detail about the user income information and guarantor. After the users have submitted it into the headquarters, it would be reviewed by the staff and the approval would be given by the higher position. The approval is given by conducting a monthly meeting between the staff and the higher position.

The current system has a lot of disadvantages, that is why we proposed a solution by making all these services online. By having it online, it could save a lot of time for the users and KADA, and it could also increase the KADA work efficiency.

1.2 The Existing System

KADA (Lembaga Kemajuan Pertanian Kemubu) is an agriculture organization in Kelantan. The workers under KADA are being given the opportunity to become cooperative members of KADA and make a loan under KADA too. But, the current system, or the as-is system in KADA to provide such services is being done manually and there is where we saw that it has very low efficiency and productivity. According to the list below, are the services that KADA provided to their staff currently.

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2. Applying a loan in cooperative of the KADA

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The current system has a lot of problem such as :

1. Loss of information

Most of the records are stored in paper such as the registration documents and also the loan documents. Paper documents can be easily damaged by water, fire or mishandling. Once damaged, the information on them may become unreadable or entirely lost. Other than that, paper documents can be misplaced and lost due to human error.

1. Accumulation of data when access or update

The clerk must locate the paper back in order to update the data, which is a difficult operation because they must scan and identify each paper in their safe in order to locate the particular one. As a result, it will take longer to finish a single task—updating the user's information

1. Highly dependent on human

Using paper forms greatly reduces productivity and efficiency because handling, filing, and finding these documents take a lot of time and effort. The dependence on paper slows down operations, causes delays, and keeps employees from focusing on more important tasks, ultimately making the organization less productive.

1. Difficulty in distributing and receiving the application

In order to obtain the form and submit it, applicants must make an extensive journey to the headquarters. They are certain to be burdened by this since they will need to find time to travel to the headquarters and spend money on transportation.

The proposed online system should replace the existing manual system because it effectively addresses critical issues such as data loss, inefficiency, human dependency, and limited accessibility. Currently, paper-based records are vulnerable to damage, loss, and misplacement, while the new system securely stores data electronically, ensuring better protection through backups and encryption. Manual processes require clerks to spend excessive time locating and updating records, whereas the digital system allows instant access and streamlined management, significantly improving productivity. Additionally, the reliance on human labor in the manual system slows operations, but automation in the proposed system reduces workload and accelerates application processing. For applicants, the current system requires physical visits to the headquarters, which is inconvenient and costly, whereas the online system allows users to submit forms remotely, anytime, with immediate confirmation, saving time and travel expenses. Members of the Board of Directors (ALK) also benefit from remote access to applications, enabling faster reviews and approvals. Overall, the proposed system enhances efficiency, accuracy, and user satisfaction, making it a superior solution that should replace the outdated manual system.

Table 1: Comparison of existing systems

<i>Features</i>	<i>Existing system (manual)</i>	<i>Proposed system (online)</i>
<i>Data Storage</i>	<i>Paper-based records prone to damage, loss, or misplacement.</i>	<i>Centralized, secure digital storage with backups and encryption.</i>
<i>Data Accessibility</i>	<i>Manual search through physical files, time-consuming and inefficient</i>	<i>Instant access to data through a searchable, user-friendly interface.</i>
<i>Application submission</i>	<i>Requires in-person visits to headquarters for form submission</i>	<i>Online form submission accessible from anywhere with an internet connection</i>

1.3 The Proposed System

The system will serve as a cutting-edge substitute for the outdated paperwork system. Thus, the system will perform the same basic tasks as the original paperwork system, including handling the registration form for the new member and also the registration form to apply for a loan for the member of the organization. On the other hand, The shift from physical documentation to an online form is the main upgrade for the system. In the past, the organization primarily employed paper in their operations to store user information and to verify users during the loan application and registration processes. Initially, the current system only had three primary user views: the applicator, Ahli Lembaga Pengarah (ALK), and the clerk. However, after a few discussions, we concluded that the admin roles would be essential for the new online system.

Firstly, from the view of the applicants, The new system will definitely be beneficial for the user as the online system offers many benefits for the applicants. With the current paper system, people have to travel to the headquarters to get and submit forms, which takes a lot of time and money. This can especially be difficult for those who live far away or have busy schedules. In contrast, the online system allows people to access and submit forms anytime and from anywhere with an internet connection. This means they don't need to make a special trip, saving both time and travel costs. The online system is available 24/7, making it much more convenient. It also provides instant confirmation when an application is submitted, so applicants don't have to worry whether their forms are received by KADA or not. Additionally, online forms often help prevent mistakes by checking for missing or incorrect information before submission. Overall, the online application system makes the process faster, easier, and cheaper for everyone involved.

Secondly, for the members of the Board of Directors (ALK), their job of checking membership and loan applications becomes much easier with the new online system. Before, this process was slow and difficult because they had to be at the office and deal with lots of paperwork, which limited their flexibility and caused delayed decisions. With the online system, ALK members can do their work from anywhere and at any time, without needing to travel or handle physical documents. This means they can approve applications faster and more easily. The system also provides instant updates and notifications, making their work more efficient. As a result, the organization becomes much more productive in completing tasks that used to take days or weeks in a much shorter time. This makes the work of ALK members easier and improves the overall efficiency and speed of the organization.

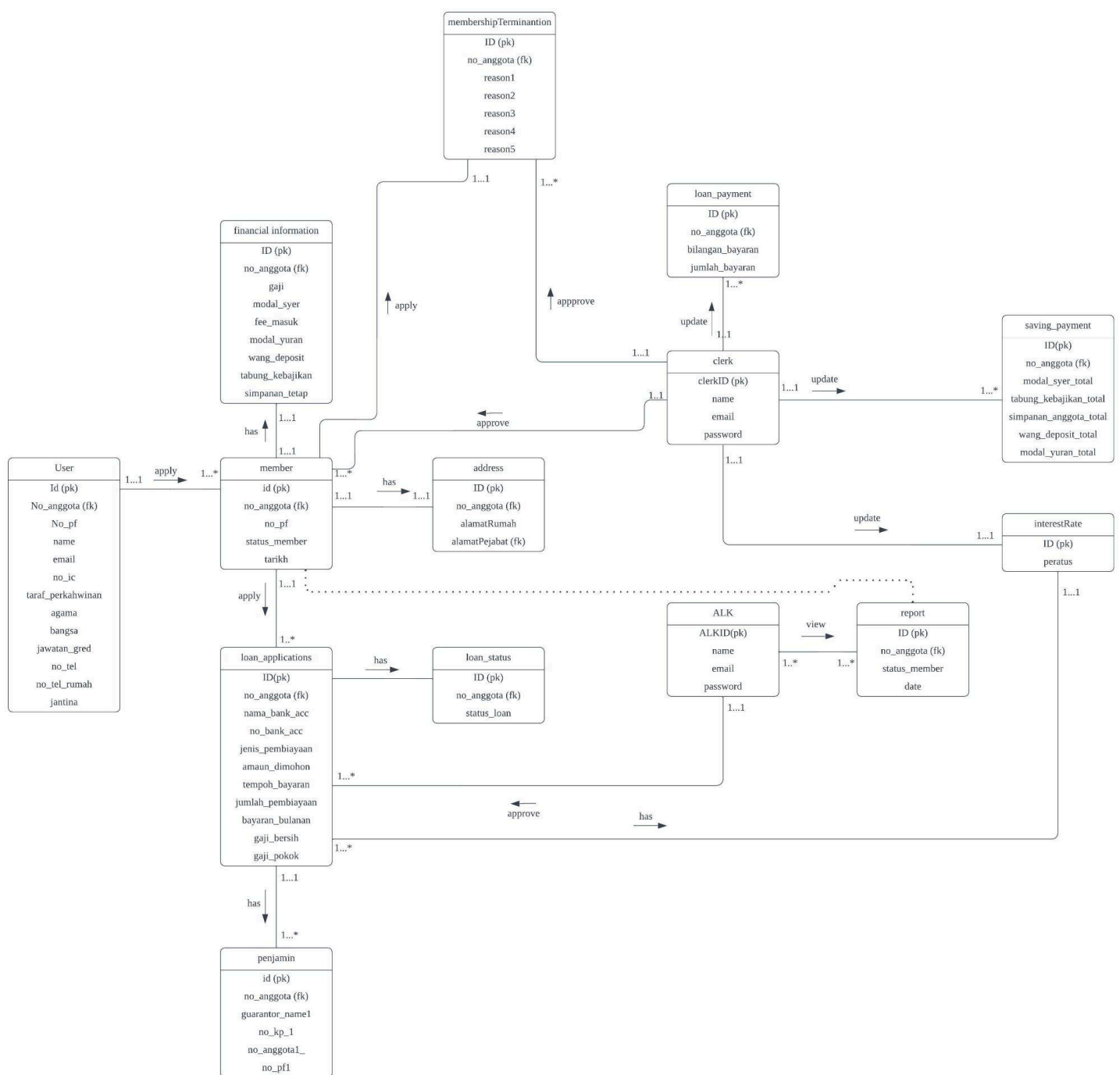
The discussion revolves around creating a user-friendly system interface for clerks to handle applications smoothly. This interface allows clerks to easily access applicant details and payment records, and to update application statuses promptly. A key task for clerks is to review submitted forms, ensuring they meet the requirements set by KADA. With the system's filtering features, clerks can quickly sort through applications, accepting those that meet criteria and rejecting incomplete or non-compliant ones. By diligently updating database records, clerks maintain accurate information for organizational use. This streamlined process helps clerks efficiently manage applications, contributing to the organization's overall effectiveness.

Lastly, for the administrative, their role in this new proposed system is to maintain the data and also the system itself. The administration needs to ensure that the system can run smoothly without any problem. Other than that, they also need to ensure that every

application information is recorded in the database so that the data is safely secure.

Global Conceptual Data Model (Global ERD)

2.1 Global Conceptual Database Design (Conceptual ERD)



2.2 Data Dictionary for the global conceptual ERD.

2.2.1 Identifying Entity

Entiti Name	Description	Aliases	Occurrence
user	Staff who do not have an active account yet.	client	Occurs when a KADA staff member decides to become the cooperative member.
member	User who already owns an account	User	occurs when a member has already log in
penjamin	Guarantor is to verified the details accuracy for member in the loan application	Guarantor	occurs when member filled in for loan application
address	The physical location associated with a member, which can include both home and office addresses.	Location, Contact Address, Residential/Work Address	Provided when a member registers or updates their profile, ensuring accurate records for communication and official purposes.
membership termination	terminate membership form for members	terminate membership form	This occurs when a member wants to leave the KADA Cooperative.
loanApplication	Loan application form for members	loan form	occurs when member want to make a loan from KADA cooperative.
loanPayment	clerk will record the amount of payment that already deducted from users salary	loan payment record	occurs when the clerk wants to record the users monthly payment
savingsPayment	A payment made by a user towards their savings account	Contribution, Savings Contribution	Occurs when a user deposits money into their savings

			account, either manually or through an automated deduction.
loanStatus	give a current loan status to the member whether it has been approved, pending or not approved	current loan state	occurs when member click on the profile
financial_information	Details about a user's financial status, including their loan preferences and repayment capability.	Financial Data, Budget Details	Occurs when a user selects a loan type and specifies the loan amount based on their financial capacity.
interestRate	clerk modify the interest rate for the loan based on economy current state	loan rate	occur when users want to filled in the loan application form
Clerk	A user who manages the loan application and membership	Admin	occur when the user login as a clerk
ALK	A user who verify the loan application and view the report	executives	occur when user login as ALK
report	give a overall report about the cooperation	summary	occur when ALK click on the report

2.2.2 Identifying Relationship Types

Entity Name	Multiplicity	Relationship	Entity Name	Multiplicity
User	1...*	Apply	member	1...1
member	1...*	Apply	loanApplication	1...1
	1..1	Edit	Profile	1...*
	1...*	Pay	Payment	1...1
	1...1	view	loanStatus	1...1
	1...1	update	homeAddress	1...1
	1...1	update	officeAddress	1...1
	1...1	update	financial_inform ation	1...1
	1...1	make	saving payment	1..1
Clerk	1...*	Manage	Loan	1...1
	1...*	approve	member	1...1
	1...*	update	member status	1...1
	1...*	update	Saving payment	1...1
	1...*	update	loanPayment	1...1
	1...*	manage	membership termination	1...1
	1...1	update	interestRate	1...1
	1...1			
ALK	1...*	Verify	LoanApplication	1...1
	1...*	view	Report	1...*

2.2.3 Attributes Description

Entity Name	Attributes	Data Type & length	Description	Nul l	Mul ti- valu ed	Example
member	no_anggota (PK)	Varchar (255)	Uniquely identify each membership number	No	No	A234
	nama	Varchar (255)	Member's full name	No	No	Nur Ain Binti Azim
	no_IC	Varchar (20)	Member identification	No	No	981223-10-32 46

			card number			
	email	Varchar (255)	Member's unique email address	No	No	naimboekk@gmail.com
	role	Varchar (50)	Member's position in KADA	No	No	Clerk
penjamin 1	nama	Varchar (255)	Guarantor's 1 full name	No	No	Aina Husna Binti Ahmad
	no_ic_p1	Varchar (20)	Guarantor's 1 identification card number	No	No	991025-14-23 42
	no_anggota_p1 (PK)	Varchar (255)	Guarantor's 1 unique membership number	No	No	A541
	no_pf1	Varchar (50)	Guarantor's 1 registration number	No	No	RG2300
penjamin 2	nama	Varchar (255)	Guarantor's 2 full name	No	No	Nisa Sofea Binti Ali
	no_ic_p2	Varchar (20)	Guarantor's 2 identification card number	No	No	990822-14-24 28
	no_anggota_p2 (PK)	Varchar (255)	Guarantor's 2 unique membership number	No	No	A539
	no_pf2	Varchar (50)	Guarantor's 2 registration number	No	No	RG2500
savingsPayment						
financial						

applyLoan	no_apply_loan (PK)	int (10)	after filled in all the details, the application will be provided with the loan application serial number	No	No	L11234
	no_anggota (FK)	Varchar (255)	unique membership number by member	No	No	A234
	no_anggota_p1 (FK)	Varchar (255)	Guarantor 1 unique membership number by penjamin 1	No	No	A541
	no_anggota_p2 (FK)	Varchar (255)	Guarantor 2 unique membership number by penjamin2	No	No	A539
	jenis_loan	Varchar (255)	Type of loan wants apply	No	No	Al-Bai
	amaun_dimohon	decimal (10,2)	Amount to borrowed from KADA	No	No	100001
	tempoh_bayaran	int (10)	Duration of the loan repayment	No	No	10
	bayaran_bulanan	decimal (10,2)	Monthly amount needs to be paid by the member	No	No	200
	surat_pengakuan	Varchar (255)	members upload the declaration letter files	No	No	SuratPengakuan.pdf
	surat_peng	Varchar	members	No	No	suratPengesahan

	esahan_majikan	(255)	upload the employer verification letter			hanMajikan.pdf
	gaji_pokok	decimal (10,2)	basic salary of member	No	No	10000
	gaji_bersih	decimal (10,2)	net salary of member	No	No	7000
	pembayaran_deposit	Varchar (255)	the deposit transaction proof files by the member	No	No	deposit.pdf
loanStatus	no_anggota (FK)	Varchar (255)	unique membership number by member	No	No	A234
	no pf	Varchar (255)	member registration number	No	No	RG1188
	statusLoan	text	loan current state	No	No	Approved
loanCalculation	no_anggota (FK)	Varchar (255)	unique membership number by member	No	No	A234
	no_loan (FK)	Varchar (10)	Unique loan number by loan	No	No	1123
	tempoh_loan	int (10)	Duration of the loan repayment	No	No	10
	bayaran_bulanan	decimal (10,2)	Monthly amount needs to be paid by the member	No	No	200
Clerk	email	varchar(2	Account email	No	No	weq123@gm

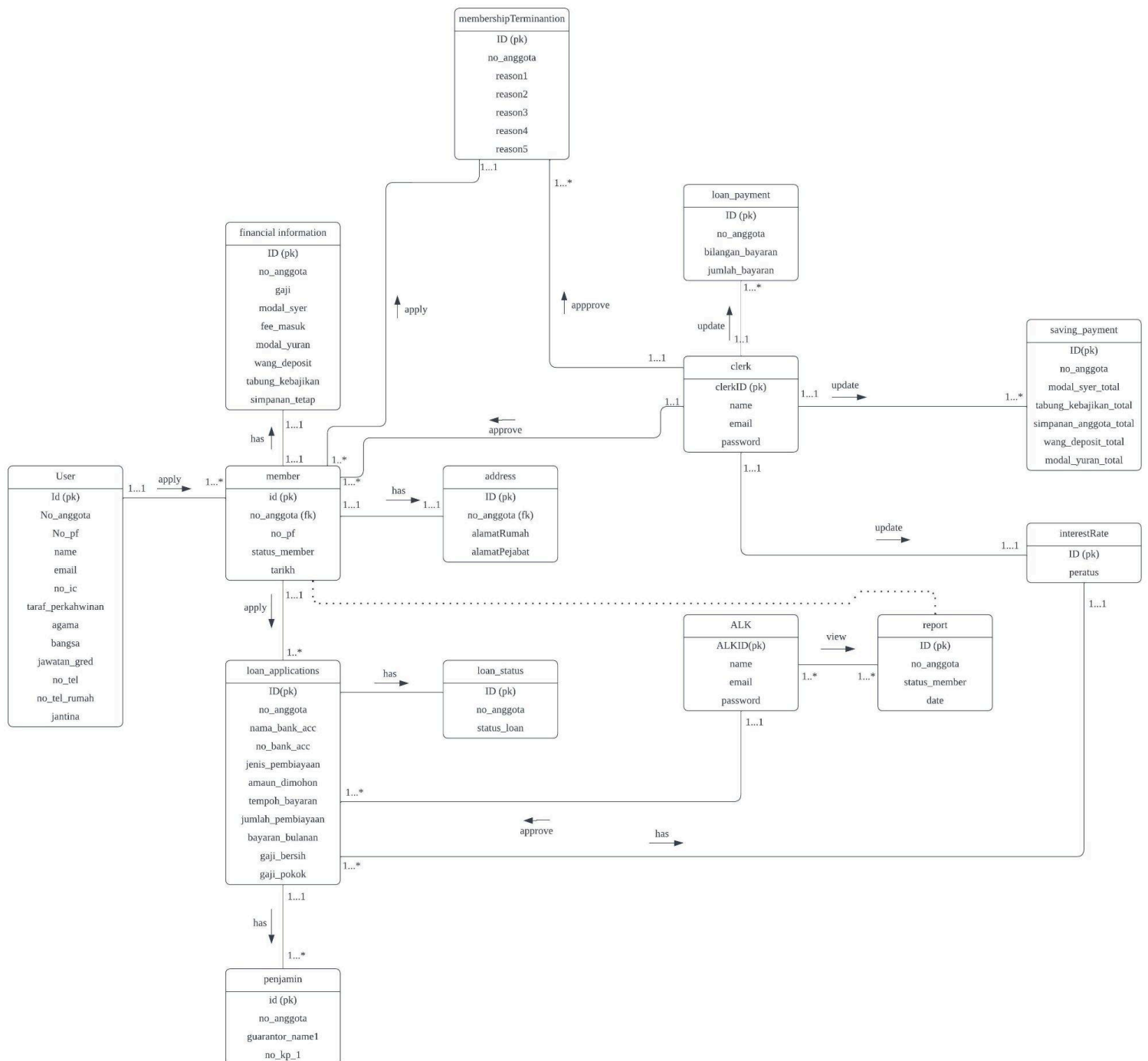
		55)	for login			ail.com
	password	varchar(255)	Account password for login	No	No	weq123@w
Account	No Anggota	varchar(255)	unique identifier for each member	No	No	A001
	Name	varchar(255)	member full name	No	No	Abu Bin Ahmad
	No IC	VARCHAR(20)	unique identifier for each member	No	No	123456-78-9012
	marriage status	varchar(255)	member's marriage status	No	No	single
	address	varchar(255)	member's home address	No	No	Universiti Teknologi Malaysia Johor Bahru 81310 UTM Johor Bahru, Johor, Malaysia.
	No PF	VARCHAR(50)	unique identifier for each member	No	No	B001
	position and Gred	VARCHAR(100)	the position and the gred of the member	No	No	officer GRED 41
	Office address	VARCHAR(100)	member's office address	No	No	Universiti Teknologi Malaysia Johor Bahru 81310 UTM Johor Bahru, Johor, Malaysia.

	monthly salary	INT	member's salary	No	No	2000
	Gender	VARCHAR (50)	member's gender	No	No	Male
	religion	VARCHAR (50)	member's religion	No	No	Islam
	race	VARCHAR (50)	member's race	No	No	Melayu
	password	varchar(255)	a unique password for the account	No	No	weq123@w
	Status	VARCHAR (50)	status of the member application	No	No	accepted
ALK	username	varchar (255)	ALK username	no	no	mukhriz Al iman
	password	varchar(255)	a uniques password for the account	no	no	123456abn.
report	tarikh	date	specific date for the report	no	no	12.3.2024
	kewangan_koperasi	int	cooperative cash flow	no	no	150000
	ahli_koperasi	int	number of the cooperative member	no	no	250
	ahli_baharu	int	number of new member of the cooperative	no	no	20
	lossProfit	int	number of profit or loss that cooperative had	no	no	15000
payment	nama	varchar(2	user full name	no	no	A001

		55)				
	no_anggota	varchar(255)	unique identifier for each member	no	no	ahmad Adib
	jumlah_bayaran	int	payment make to the system	no	no	2000
	bukti_bayaran	Varchar (255)	user upload the transaction receipt	no	no	receipt.pdf

Logical Data Model (Logical ERD)

3.1 Logical Database Design (Logical ERD)



3.2 Data Dictionary for the Logical ERD

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penjamin	Guarantor is to verified the details accuracy for member in the loan application	Guarantor	occurs when member filled in for loan application
address	The physical location associated with a member, which can include both home and office addresses.	Location, Contact Address, Residential/Work Address	Provided when a member registers or updates their profile, ensuring accurate records for communication and official purposes.
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savingsPayment	A payment made by a user towards their	Contribution, Savings	Occurs when a user deposits money into

	savings account	Contribution	their savings account, either manually or through an automated deduction.
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User	1...*	Apply	member	1...1
member	1...* 1...1 1...1 1...1 1...1	Apply view update update make	loan_application loan_status address financial_inform ation saving payment	1...1 1...1 1...1 1...1 1..1
Clerk	1...* 1...* 1...* 1...* 1...* 1...1	Manage approve update update manage update	Loan member Saving payment loan_payment membership termination interestRate	1...1 1...1 1...1 1...1 1...1 1...1
ALK	1...* 1...*	Verify view	LoanApplication Report	1...1 1...*

3.2.3 Attributes Description

Entity Name	Attributes	Data Type & length	Description	Null	Multi-valued	Example
user	ID (PK)	Int	unique ID number (Auto increment)	No	No	1
	no_anggota (FK)	Varchar (255)	unique staff number	No	No	A541
	no_PF	Varchar (50)	staff personal file number	No	No	RG2355
	name	varchar(255)	name of the staff	No	No	Darwish Aliff bin Hairulnizam
	email	varchar(255)	Account email for login	No	No	qwerty123@gmail.com
	IC_no	varchar(12)	User's identification card number	no	no	040508011216
	jantina	ENUM ('Lelaki', 'Perempuan')	User's gender	no	no	Male
	taraf_perkahwinan	ENUM ('Bujang', 'Berkahwin', 'Bercerai', 'Balu')	User's marital status	no	no	bujang
	agama	VARCHAR (50)	User's religion	no	no	Islam
	bangsa	VARCHAR (50)	User's ethnicity	no	no	Melayu

	jawatan_gr ed	VARCHAR (50)	User's job grade or position	no	no	CEO Gred 41
	no_tel	Varchar(1 5)	User's personal phone number	no	no	+6012345678 9
	no_tel_rum ah	Varchar(1 5)	User's home phone number	no	no	03-4567890
member	ID (PK)	Varchar (255)	unique ID number (Auto increment)	No	No	1
	no_anggota (FK)	Varchar (255)	unique staff number	No	No	A541
	no_PF	Varchar (50)	Guarantor's 1 personal file number	No	No	RG2300
	status_me mber	Varchar (255)	status member	No	No	Accepted
penjamin	ID (PK)	Int	unique ID number (Auto increment)	No	No	1
	no_anggota (FK)	Varchar (255)	unique staff number	No	No	A541
	guarantor_ name	Varchar (255)	Guarantor's 1 name	No	No	Aina Husna Binti Ahmad
	No_KP	Varchar (20)	Guarantor's 1 identification card number	No	No	991025-14-23 42
	no_anggota	Varchar (255)	Guarantor's 1 unique staff number	No	No	A541

	no_PF	Varchar (50)	Guarantor's 1 personal file number	No	No	RG2300
address	ID (PK)	int (10)	unique ID number for each member	no	no	2
	no_anggota (FK)	Varchar (255)	unique staff number	no	no	A009
	alamatRumah	Varchar (255)	staff's home address	no	no	No 64, Jalan Pulau Perdana 4/4, Taman Sri Pulau Perdana
	alamatPejabat (FK)	Varchar (255)	staff's office address	no	no	Lembaga Kemajuan Pertanian Kemubu Peti Surat 127, Jalan Dato'Lundang 15710 Kota Bharu, Kelantan.
loan_application	ID (PK)	int (10)	unique ID number for each member (Auto increment)	No	No	5
	no_anggota (FK)	Varchar (255)	unique staff number	No	No	A005
	nama_bank_acc	Varchar (255)	members account bank name	No	No	Maybank
	no_bank_acc	Varchar (255)	members account bank number	No	No	8879 1234 8765 6754
	jenis_pembiayaan	Varchar (255)	Type of loan wants apply	No	No	Al-Bai
	jumlah_dibayarkan	decimal (10,2)	Amount to borrowed from	No	No	10000

			KADA			
	tempoh_bayaran	int (10)	Duration of the loan repayment	No	No	6
	jumlah_pembiayaan	decimal (10,2)	total loan repayment including the interest rate	No	No	1048.08
	bayaran_bulanan	decimal (10,2)	Monthly amount needs to be paid by the member	No	No	45.17
	gaji_pokok	decimal (10,2)	basic salary of member	No	No	10000
	gaji_bersih	decimal (10,2)	net salary of member	No	No	7000
loan_status	ID (PK)	int (10)	unique ID number for each status (Auto increment)	No	No	5
	no_anggota (FK)	Varchar (255)	unique staff number	No	No	A234
	status_loan	text	loan application current state	No	No	Approved
loan_payment	ID (PK)	int (10)	unique ID number for each payment (Auto increment)	No	No	5
	no_anggota (FK)	Varchar (255)	unique staff number	No	No	A005
	bilangan_bayaran	int (10)	the number of times member has already paid	No	No	12

	jumlah_bayaran	decimal (10,2)	The cumulative amount members already paid	No	No	255.85
Saving Payment	ID (PK)	Int	unique ID number (Auto increment)	No	No	1
	no_anggota (FK)	Varchar (255)	unique staff number	No	No	A541
	modal_syer_total	Decimal(10,2)	Total share capital deposited by the member	No	No	5
	tabung_kebajikan_total	Decimal(10,2)	Total welfare fund deposited by the member	No	No	5
	simpanan_anggota_total	Decimal(10,2)	Total member's savings contributed by the member	No	No	5
	wang_deposit_total	Decimal(10,2)	Total amount deposited by the member	No	No	5
	modal_yuran_total	Decimal(10,2)	Total membership fee or capital contributed by the member	No	No	35
financial_information	ID (PK)	Int	unique ID number (Auto increment)	No	No	1
	no_anggota (FK)	Varchar (255)	unique staff number	No	No	A541
	gaji	Decimal (10,2)	Monthly salary or earnings of the member	No	No	10000

	modal_syer	Decimal (10,2)	Share capital contributed by the member	No	No	5
	fee_masuk	Decimal (10,2)	Entry fee or registration fee for membership	No	No	50
	modal_yuran	Decimal (10,2)	Membership fee or capital contribution by the member	No	No	35
	wang_deposit	Decimal (10,2)	Total deposit amount held by the member	No	No	5
	tabung_kebajikan	Decimal (10,2)	Welfare fund contributed by the member	No	No	5
	simpanan_tetap	Decimal (10,2)	Fixed savings deposited by the member for a specified period	No	No	5
faedah	ID (PK)	int (10)	unique ID number for each interest rate (Auto increment)	No	No	5
	peratus	int (10)	clerk set the interest rate percentage	No	No	4.2
Clerk	ClerkID (PK)	Int	unique ID number for each Clerk (Auto increment)	No	No	1
	name	varchar(255)	name of the clerk	No	No	Aina Husna Binti Ahmad
	email	varchar(255)	Account email	No	No	weq123@gm

		55)	for login			ail.com
	password	varchar(255)	Account password for login	No	No	weq123@w
ALK	ALKID (PK)	Int	unique ID number for each ALK (Auto increment)	No	No	1
	name	varchar(255)	name of the ALK	No	No	Aina Husna Binti Ahmad
	email	varchar(255)	Account email for login	No	No	weq123@gmail.com
	password	varchar(255)	Account password for login	No	No	weq123@w
report	ID (PK)	Int	unique ID number for each report (Auto increment)	No	No	1
	no_anggota (FK)	Varchar (255)	unique membership number	No	No	A541
	status_member	Varchar(255)	member status	no	no	Accepted
	date	date	specific date for the report	no	no	12.3.2024
membership Termination	ID (PK)	Int	unique ID number (Auto increment)	No	No	1
	no_anggota (FK)	Varchar (255)	unique membership number	No	No	asd

	reason1	Varchar(255)	Reasons for wanting to terminate membership	No	No	"i want to quit"
	reason2	Varchar(255)	Reasons for wanting to terminate membership	Yes	No	"i want to quit"
	reason3	Varchar(255)	Reasons for wanting to terminate membership	Yes	No	"i want to quit"
	reason4	Varchar(255)	Reasons for wanting to terminate membership	Yes	No	"i want to quit"
	reason5	Varchar(255)	Reasons for wanting to terminate membership	Yes	No	"i want to quit"

3.3 Relational Database Schema(s)

User (ID (PK), no_anggota (FK), no_PF, name, email, IC_no, jantina, taraf_perkahwinan, agama, bangsa, jawatan_gred, no_tel, no_tel_rumah)

PK : ID

FK : no_anggota reference to member (no_anggota)

Member (ID (PK), no_anggota , no_PF, status_member)

PK : ID

financial_information (ID , no_anggota, gaji, modal_syer, fee_masuk, modal_yuran , wang_deposit, tabung_kebajikan, simpanan_tetap)

PK : ID

FK : no_anggota reference to member (no_anggota)

Address (ID ,no_anggota ,alamatRumah ,alamatPejabat)

PK : ID

FK: no_anggota reference to member (no_anggota)

membership_termination (ID, no_anggota, reason1, reason2, reason3, reason4, reason5)

PK : ID

FK: no_anggota reference to member (no_anggota)

Clerk (ClerkID, name , email, password)

PK ; ClerkID

loan_application (ID, no_anggota, nama_bank_acc, no_bank_acc, jenis_pembiayaan, amaun_dimohon, tempoh_bayaran, jumlah_pembiayaan, bayaran_bulanan, gaji_bersih, gaji_pokok)

PK: ID

FK: no_anggota reference to member (no_anggota)

loan_status (ID, no_anggota, status_loan)

PK: ID

FK: no_anggota reference to member (no_anggota)

Penjamin (ID, no_anggota, guarantor_name1, No_KP_1, no_anggota_1, no_PF1, guarantor_name2, No_KP_2, no_anggota_2, no_PF2)

PK : ID

loan_payment (ID, no_anggota, bilangan_bayaran, jumlah_bayaran)

PK: ID

FK: no_anggota reference to member (no_anggota)

Saving_Payment (ID , no_anggota, modal_syer_total, tabung_kebajikan_total , simpanan_anggota_total , wang_deposit_total , modal_yuran_total)

PK : ID

FK: no_anggota reference to member (no_anggota)

faedah (ID , peratus)

PK : ID

ALK (ALKID, name, email, password)

PK : ALKID

Report (ID, no_anggota, status_member, date)

PK : ID

3.4 Normalization

1NF

User (ID (PK), no_anggota (FK), no_PF, name, email, IC_no, jantina, taraf_perkahwinan, agama, bangsa, jawatan_gred, no_tel, no_tel_rumah)

PK : ID

FK : no_anggota reference to member (no_anggota)

Member (ID (PK), no_anggota , no_PF, status_member, tarikh)

PK : ID

financial_information (ID , no_anggota, gaji, modal_syer, fee_masuk, modal_yuran , wang_deposit, tabung_kebajikan, simpanan_tetap)

PK : ID

FK : no_anggota reference to member (no_anggota)

Address (ID ,no_anggota ,alamatRumah ,alamatPejabat)

PK : ID

FK: no_anggota reference to member (no_anggota)

membership_termination (ID, no_anggota, reason1, reason2, reason3, reason4, reason5)

PK : ID

FK: no_anggota reference to member (no_anggota)

Clerk (ClerkID, name , email, password)

PK ; ClerkID

loan_application (ID, no_anggota, nama_bank_acc, no_bank_acc, jenis_pembiayaan, amaun_dimohon, tempoh_bayaran, jumlah_pembiayaan, bayaran_bulanan, gaji_bersih, gaji_pokok)

PK: ID

FK: no_anggota reference to member (no_anggota)

loan_status (ID, no_anggota, status_loan)

PK: ID

FK: no_anggota reference to member (no_anggota)

Penjamin (ID, no_anggota, guarantor_name, No_KP, no_anggota, no_PF)

PK : ID

FK: no_anggota reference to member (no_anggota)

loan_payment (ID, no_anggota, bilangan_bayaran, jumlah_bayaran)

PK: ID

FK: no_anggota reference to member (no_anggota)

Saving_Payment (ID , no_anggota, modal_syer_total, tabung_kebajikan_total , simpanan_anggota_total , wang_deposit_total , modal_yuran_total)

PK : ID

FK: no_anggota reference to member (no_anggota)

faedah (ID , peratus)

PK : ID

ALK (ALKID, name, email, password)

PK : ALKID

Report (ID, no_anggota, status_member, date)

PK : ID

2NF

There are no partial dependencies in the derived relations, therefore they are already in Second Normal Form (2NF). As a result, all derived dependencies remain unchanged except the Benjamin table , matching with the First Normal Form.

3NF

Since there are no partial or transitive dependencies in any of the tables, they are already in 3NF. Thus, no further normalization is required, and the table satisfies 3NF directly without additional processing.

Interface Design and SQL Implementation

4.1 Interface Design (Mapped to the Proposed SQL Statement)

User Module

- 1) Interface to apply for membership

The screenshot shows a web form titled "Permohonan Menjadi Ahli" (Membership Application). The form is divided into a header section and a main content area. The header section has a green background with the title "Permohonan Menjadi Ahli" in white. The main content area has a white background with a green border. The form is titled "Maklumat Peribadi" (Personal Information) in green. It contains several input fields: "Nama:" (Name) with the value "YASMIN BATRISYIA BINTI ZAHIRUDDIN", "Kad Pengenalan:" (Identification Card) with the value "040124140466", "Jantina:" (Gender) with a dropdown menu showing "Perempuan", "Agama:" (Religion) with a dropdown menu showing "Islam", "Bangsa:" (Race) with a dropdown menu showing "Melayu", "Taraf Perkahwinan:" (Marital Status) with a dropdown menu showing "bujang", and "Gaji:" (Salary) with the value "10000". There is a blue button labeled "Seterusnya" (Next) at the bottom.

The screenshot shows a web form titled "Permohonan Menjadi Ahli" (Membership Application). The form is divided into a header section and a main content area. The header section has a green background with the title "Permohonan Menjadi Ahli" in white. The main content area has a white background with a green border. The form is titled "Maklumat Alamat" (Address Information) in green. It contains several input fields: "Alamat Rumah:" (Home Address) with the value "22, Elmina, Shah Alam, Selangor", "Poskod:" (Postcode) with the value "40140", and "Negeri:" (State) with a dropdown menu showing "Selangor". There are two buttons at the bottom: a grey button labeled "Sebelum" (Previous) and a blue button labeled "Seterusnya" (Next).

Permohonan Menjadi Ahli

Maklumat Pekerjaan

No Anggota: A007	No PF: 0809
Jawatan & Gred: Pegawai Kanan	Alamat Pejabat: KADA, Kota Bharu, Kelantan

Sebelum
Seterusnya

Permohonan Menjadi Ahli

Maklumat Perhubungan

Emel: ybzahiruddin@gmail.com	No Tel Bimbit: 012-2234456 Format: 012-3456789
No Tel Rumah: 03-23456789 Format: 03-12345678	

Sebelum
Seterusnya

```

INSERT INTO Member (ID, no_anggota, no_PF, status_member)
VALUES (
    '${MemberData["id"]}',
    '${MemberData["no_anggota"]}',
    '${MemberData["no_PF"]}',
    '${MemberData["status_member"]}'
);

```

```

INSERT INTO User (ID, no_anggota, no_PF, name, email, IC_no, jantina,
taraf_perkahwinan, agama, bangsa, jawatan_gred, no_tel, no_tel_rumah)
VALUES (
    '{$UserData["id"]}',
    '{$UserData["no_anggota"]}',
    '{$UserData["no_PF"]}',
    '{$UserData["name"]}',
    '{$UserData["email"]}',
    '{$UserData["IC_no"]}',
    '{$UserData["jantina"]}',
    '{$UserData["taraf_perkahwinan"]}',
    '{$UserData["agama"]}',
    '{$UserData["bangsa"]}',
    '{$UserData["jawatan_gred"]}',
    '{$UserData["no_tel"]}',
    '{$UserData["no_tel_rumah"]}'
);

```

```

INSERT INTO financial_information (ID, no_anggota, gaji, modal_syer, fee_masuk,
modal_yuran, wang_deposit, tabung_kebajikan, simpanan_tetap)
VALUES (
    '{$FinanceData["id"]}',
    '{$FinanceData["no_anggota"]}',
    '{$FinanceData["gaji"]}',
    '{$FinanceData["modal_syer"]}',
    '{$FinanceData["fee_masuk"]}',
    '{$FinanceData["modal_yuran"]}',
    '{$FinanceData["wang_deposit"]}',
    '{$FinanceData["tabung_kebajikan"]}',
    '{$FinanceData["simpanan_tetap"]}'
);

```

```

INSERT INTO Address (ID, no_anggota, alamatRumah, alamatPejabat)
VALUES (

```

```
'{$AddressData["id"]}',  
'{$AddressData["no_anggota"]}',  
'{$AddressData["alamatRumah"]}',  
'{$AddressData["alamatPejabat"]}'  
);
```

2) Profile

PROFIL

Maklumat Peribadi

Nama:

SYARIFAH DANIA

Kad Pengenalan:

040508011218

Jantina:

Lelaki

Agama:

Islam

Bangsa:

Melayu

Taraf Perkahwinan:

Bujang

Gaji:

1222.00

Maklumat Alamat

Alamat Rumah:

NO 64 JALAN PULAI PERDANA 4/5 TAMAN SRI PULAI PERDANA

Maklumat Pekerjaan

No Anggota:

123

No PF:

458

Jawatan & Gred:

BOSS

Alamat Pejabat:

KOTA ISKANDAR, ISKANDAR PUTERI

Maklumat Perhubungan

Emel:

syarifahdania8504@gmail.com

No Tel Bimbit:

0187732577

No Tel Rumah:

0164582499

Change Password

Current Password:

New Password:

Confirm New Password:

Hantar

SELECT

```
u.name AS "Nama",  
u.IC_no AS "Kad Pengenalan",  
u.jantina AS "Jantina",  
u.agama AS "Agama",  
u.bangsa AS "Bangsa",
```

u.taraf_perkahwinan AS "Taraf Perkahwinan",
f.gaji AS "Gaji",
u.email AS "Emel",
u.no_tel AS "No Tel Bimbit",
u.no_tel_rumah AS "No Tel Rumah",

m.no_anggota AS "No Anggota",
m.no_PF AS "No PF",
u.jawatan_gred AS "Jawatan & Gred",

a.alamatPejabat AS "Alamat Pejabat",
a.alamatRumah AS "Alamat Rumah"

FROM

User u

JOIN

Member m ON u.no_anggota = m.no_anggota

JOIN

Address a ON m.no_anggota = a.no_anggota

JOIN

financial_information f ON m.no_anggota = f.no_anggota

WHERE

u.IC_no = '{IC_no}'

OR m.no_anggota = '{no_anggota}'

OR u.email = '{email}';

Loan Module

1) Apply loan

Permohonan Pembiayaan Anggota

Maklumat Peribadi

Nama:	Kad Pengenalan:
NAIM DANIEL	040116011729
E-mel:	Tarikh Lahir:
imandyyan@gmail.com	dd/mm/yyyy 
Jantina:	Agama:
Lelaki	Islam
Bangsa:	
India	

Permohonan Pembiayaan Anggota

Maklumat Alamat

Alamat Rumah:	Poskod:
kl	123
Negeri:	
Wilayah Persekutuan	

Permohonan Pembiayaan Anggota

Maklumat Pekerjaan

No Anggota:

A001

No PF:

1234

Jawatan & Gred:

pembantu n28

Alamat Pejabat:

kl

Poskod:

123

Sebelum

Seterusnya

Permohonan Pembiayaan Anggota

Maklumat Kewangan

Nama Bank Akaun:

rhb

Nombor Akaun Bank:

123

Sebelum

Seterusnya

Permohonan Pembiayaan Anggota

Butir-butir Pembiayaan

Jenis Pembiayaan:

Al-Innah

Kadar Faedah:

2.1

Amaun Dimohon

RM 10000

Tempoh Bayaran (Tahun):

2

Bayaran Bulanan:

RM 434.17

Jumlah Bayaran Pembiayaan:

RM 10420.00

Borang Pengakuan Pemohon:

Muat Turun Pengakuan Pemohon

Muat Naik Borang Pengakuan:

Choose File

Database E...s foot) (1).jpeg

Sebelum

Seterusnya

Permohonan Pembiayaan Anggota

Butir-butir Penjamin

Penjamin 1

Nama Penuh:

mukhriz

Nombor Kad Pengenalan:

123

No Anggota:

123

No PF:

123

Penjamin 2

Nama Penuh:

naim

Nombor Kad Pengenalan:

321

No Anggota:

321

No PF:

321

Sebelum

Seterusnya

Permohonan Pembiayaan Anggota

Pengesahan Majikan

Borang Pengesahan Majikan:

Muat Turun Borang Pengesahan Majikan

Muat Naik Surat Pengesahan Majikan

Choose File

Database ER dia...ow's foot) (1).jpeg

Gaji Pokok Sebulan

RM 10000

Gaji Bersih Sebulan

RM 10000

Sebelum

Hantar

```
INSERT INTO loan_application (ID, no_anggota, nama_bank_acc, no_bank_acc,
jenis_pembiayaan, amaun_dimohon, tempoh_bayaran, jumlah_pembiayaan,
bayaran_bulanan, gaji_bersih, gaji_pokok)
```

```
VALUES ('{$LoanData["id"]}', '{$LoanData["no_anggota"]}',
'{$LoanData["nama_bank_acc"]}', '{$LoanData["no_bank_acc"]}',
```

```

'{$LoanData["jenis_pembiayaan"]}',      '{$LoanData["amaun_dimohon"]}',
'{$LoanData["tempoh_bayaran"]}',        '{$LoanData["jumlah_pembiayaan"]}',
'{$LoanData["bayaran_bulanan"]}',      '{$LoanData["gaji_bersih"]}',
'{$LoanData["gaji_pokok"]}') );

```

```

INSERT INTO loan_status (ID, no_anggota, status_loan)

```

```

VALUES      ('{$LoanStatusData["id"]}',      '{$LoanStatusData["no_anggota"]}',
'{$LoanStatusData["status_loan"]}');

```

```

INSERT INTO loan_payment (ID, no_anggota, bilangan_bayaran, jumlah_bayaran)

```

```

VALUES      ('{$LoanPaymentData["id"]}',      '{$LoanPaymentData["no_anggota"]}',
'{$LoanPaymentData["bilangan_bayaran"]}',
'{$LoanPaymentData["jumlah_bayaran"]}');

```

2) Check loan status

Status Permohonan Pembiayaan			
Nama	No. Anggota	Status	Tindakan
NAIM DANIEL	A001	pending	Maklumat lanjut...

```

SELECT

```

```

loan_application.ID AS loan_app_id,
loan_application.no_anggota,
loan_application.nama_bank_acc,
loan_application.no_bank_acc,
loan_application.jenis_pembiayaan,
loan_application.amaun_dimohon,
loan_application.tempoh_bayaran,
loan_application.jumlah_pembiayaan,
loan_application.bayaran_bulanan,
loan_application.gaji_bersih,

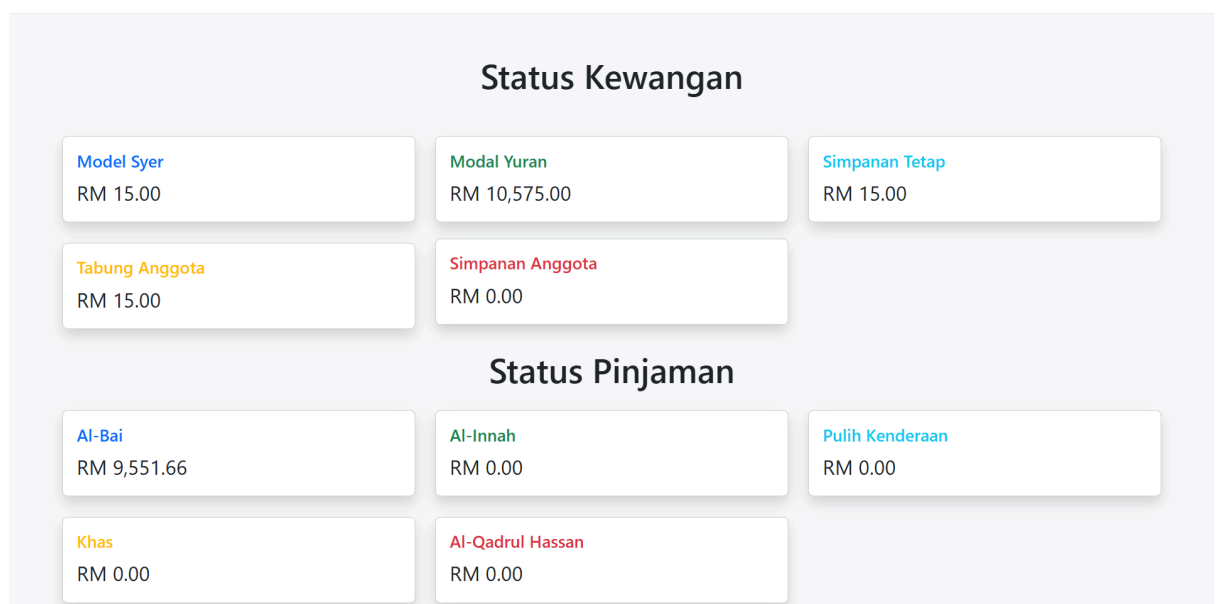
```

```

loan_application.gaji_pokok,
loan_status.ID AS loan_status_id,
loan_status.status_loan
FROM loan_application
LEFT JOIN loan_status ON loan_application.no_anggota = loan_status.no_anggota;

```

3) Status kewangan



SELECT

sp.modal_syer_total AS "Modal Syer",
sp.modal_yuran_total AS "Modal Yuran",
sp.simpanan_anggota_total AS "Simpanan Anggota",
sp.tabung_kebajikan_total AS "Tabung Anggota",
sp.wang_deposit_total AS "Simpanan Tetap",
lp1.jumlah_bayaran AS "Al-Bai",
lp2.jumlah_bayaran AS "Al-Innah",
lp3.jumlah_bayaran AS "Pulih Kenderaan",
lp4.jumlah_bayaran AS "Khas",
lp5.jumlah_bayaran AS "Al-Qadrul Hassan"

FROM

Saving_Payment sp

LEFT JOIN loan_payment lp1 ON sp.no_anggota = lp1.no_anggota AND lp1.loan_type = 'Al-Bai'

LEFT JOIN loan_payment lp2 ON sp.no_anggota = lp2.no_anggota AND lp2.loan_type = 'Al-Innah'

LEFT JOIN loan_payment lp3 ON sp.no_anggota = lp3.no_anggota AND lp3.loan_type = 'Pulih Kenderaan'

LEFT JOIN loan_payment lp4 ON sp.no_anggota = lp4.no_anggota AND lp4.loan_type = 'Khas'

LEFT JOIN loan_payment lp5 ON sp.no_anggota = lp5.no_anggota AND lp5.loan_type = 'Al-Qadrul Hassan'

WHERE

sp.no_anggota = '{no_anggota}';

Clerk Module

1) List of membership application



Nama	Kad Pengenalan	Tindakan
YASMIN BATRISYIA BINTI ZAHIRUDDIN	040124140466	More Accept Reject

[← Kembali](#)

```
SELECT
    m.m_app_id,
    m.m_user_id,
    u.IC_no,
    u.name,
    m.status_member
FROM
    membership m
JOIN
    user u ON m.m_user_id = u.ID;
```

– To accept membership

```
UPDATE membership SET m_status = 'Accepted' WHERE id = :id;
```

–To reject

```
UPDATE membership SET m_status = 'Rejected' WHERE m_app_id = :id;
```

2) List of requested loan application

Pemohonan Pembiayaan			
Name	Kad Pengenalan	tindakan	nota
YASMIN BATRISYIA BINTI ZAHIRUDDIN	040124140466	maklumat	nota
← Kembali			

```
SELECT la.ID, la.no_anggota, la.nama_bank_acc, la.no_bank_acc,
la.jenis_pembiayaan, la.amaun_dimohon, la.tempoh_bayaran, la.jumlah_pembiayaan,
la.bayaran_bulanan, la.gaji_bersih, la.gaji_pokok
```

```
FROM loan_application la
```

```
JOIN loan_status ls ON la.ID = ls.ID
```

```
WHERE ls.status_loan = 'Pending';
```

3) List of KADA Cooperative members

Senarai Ahli	
No. anggota	tindakan
A009	More
A007	More
← Kembali	

```
SELECT m.no_anggota, u.name, u.email, u.no_tel, u.jawatan_gred
```

```
FROM Member m
```

```
JOIN User u ON m.no_anggota = u.no_anggota
```

```
JOIN membership_termination mt ON m.no_anggota = mt.no_anggota
```

```
WHERE mt.status_member != 'Quit';
```

4) Cooperative member details

Detail Pemohonan Ahli

Name	YASMIN BATRISYIA BINTI ZAHIRUDDIN
Kad Pengenalan	040124140466
Jantina	Perempuan
Agama	Islam
Bangsa	Melayu
Taraf Perkahwinan	bujang
Alamat Rumah	22, Elmina, Shah Alam, Selangor
Poskod	40140
Negeri	Selangor
No Anggota	A007
No PF	0809
Jawatan & Gred	Pegawai Kanan
Alamat Pejabat	KADA, Kota Bharu, Kelantan
emel	ybzahiruddin@gmail.com
No Tel Bimbit	012-2234456
No Tel Rumah	03-23456789
Gaji	10000.00
Fee Masuk	50.00
Modal Yuran	35.00
Modal Syer	5.00
Wang Deposit	5.00
Tabung Kebajikan	5.00
Simpanan Tetap	5.00

```
SELECT m.no_anggota,m.name, u.no_IC, u.jantina, u. agama, u.bangsa, u.
taraf_perkahwinan, u.alamat_rumah, u.poskod, u.negeri, u.no_PF, u.email, u.no_tel,
u.jawatan_gred, fi.gaji, fi.modal_syer, fi.fee_masuk, fi.modal_yuran, fi.wang_deposit,
fi.tabung_kebajikan, fi.simpanan_tetap
```

```
FROM Member m
```

```
JOIN User u ON m.no_anggota = u.no_anggota
```

```
JOIN financial_information fi ON m.no_anggota = fi.no_anggota;
```

5) List of loan applicants

Senarai Pembiayaan		
Name	Kad Pengenal	tindakan
SYARIFAH DANIA	040508012122	more Delete

[← Kembali](#)

-- Select loan applicants who are pending

```
SELECT    la.ID,      la.no_anggota,    la.nama_bank_acc,    la.no_bank_acc,
la.jenis_pembiayaan, la.amaun_dimohon, la.tempoh_bayaran, la.jumlah_pembiayaan

FROM loan_application la

JOIN loan_status ls ON la.ID = ls.ID

WHERE ls.status_loan = 'Pending';
```

-- Delete loan applicants who have finished paying (assuming 'loan_payment' has a status or flag for fully paid)

```
DELETE FROM loan_application

WHERE ID IN (

    SELECT la.ID

    FROM loan_application la

    JOIN loan_payment lp ON la.no_anggota = lp.no_anggota

    WHERE lp.bilangan_bayaran = lp.jumlah_bayaran -- or any other condition that
indicates full payment

);
```

6) Loan applicant details

Detail Pemohonan Pembiayaan

Name	SYARIFAH DANIA
Email	isaamdsh@gmail.com
Kad Pengenalan	040508012122
Tarikh Lahir	2025-02-18
Jantina	Perempuan
Agama	Kristian
Bangsa	Cina
Alamat Rumah	ddwd
Poskod	21345
Negeri	Johor
No Anggota	A009
No PF	777
Jawatan/Gred	PEAUSEIF
Alamat Pejabat	CSC*
Poskod Pejabat	123
Nama Bank Akaun	rhb
No Bank Akaun	123
Jenis Pembiayaan	Al Bai
Amaun Dimohon	200.00
Tempoh Bayaran	2
Bayaran Bulanan	9.03
Nama Penjamin 1	mukhriz
No KP Penjamin 1	123
No Anggota Penjamin 1	A006
No PF Penjamin 1	123
Nama Penjamin 2	naim
No KP Penjamin 2	321
No Anggota Penjamin 2	A005
No PF Penjamin 2	321
Gaji Pokok	250.00
Gaji Bersih	260.00
Borang Pengakuan Pemohon	View Document
Borang Pengesahan Majikan	View Document

```
SELECT u.name, u.email, u.IC_no, u.birthday, u.jantina, u.agama, u.bangsa,
a.alamatRumah, a.poskod, a.negeri, m.no_anggota, m.no_PF, m.jawatan_gred,
a.alamatPejabat, la.nama_bank_acc, la.no_bank_acc, la.jenis_pembiayaan,
la.amaun_dimohon, la.tempoh_bayaran, la.jumlah_pembiayaan, la.bayaran_bulanan,
la.gaji_bersih, la.gaji_pokok
```

```
FROM Member m
```

```
JOIN User u ON m.no_anggota = u.no_anggota
```

```
JOIN loan_application la ON m.no_anggota = la.no_anggota
```

```
JOIN Address a ON m.no_anggota = a.no_anggota
```

WHERE la.ID IS NOT NULL;

7) Interest rate

Kadar Faedah

Peratus:

%

Kemaskini

← Kembali

```
SELECT
    f.ID AS "ID",
    f.peratus AS "Peratus"
FROM
    faedah f
WHERE
    f.ID = '{ID}';
```

```
UPDATE faedah
SET peratus = '{peratus}'
WHERE ID = '{ID}';
```

8) Saving Payment

 Home Senarai Ahli Senarai Pembiayaan Pemohonan Ahli Pemohonan Pembiayaan						
Pembayaran Simpanan						
No Anggota	Jenis Pembiayaan	Tempoh Bayaran (Tahun)	Bilangan Pembayaran	Jumlah Bayaran Belum Dibayar	Jumlah Bayaran Dah Dibayar	Tindakan
A007	Al_Bai	2	0	1084	0.00	Kemaskini
← Kembali						

SELECT

u.name,

u.email,

sp.no_anggota,

sp.modal_syer_total,

sp.tabung_kebajikan_total,

sp.simpanan_anggota_total,

sp.wang_deposit_total,

sp.modal_yuran_total

FROM

Saving_Payment sp

JOIN

Member m ON sp.no_anggota = m.no_anggota

JOIN

User u ON m.no_anggota = u.no_anggota;

9) Update saving payment

Butiran Pembayaran

Modal Syer	5.00
Modal Yuran	3525.00
Wang Deposit	5.00
Tabung Kebajikan	5.00
Simpanan Tetap	5.00
Kemaskini	

Jumlah Terkini

Jenis	Jumlah
Modal Syer Total	0
Modal Yuran Total	0
Wang Deposit Total	0
Tabung Kebajikan Total	0
Simpanan Tetap Total	0

UPDATE Saving_Payment

SET modal_syer_total = {modal_syer_total},

tabung_kebajikan_total = {tabung_kebajikan_total},

simpanan_anggota_total = {simpanan_anggota_total},

wang_deposit_total = {wang_deposit_total},

modal_yuran_total = {modal_yuran_total}

WHERE no_anggota = no_anggota;

10) Loan Payment

							Home	Senarai Ahli	Senarai Pembiayaan	Pemohonan Ahli	Pemohonan Pembiayaan
Pembayaran Simpanan											
No Anggota	Jenis Pembiayaan	Tempoh Bayaran (Tahun)	Bilangan Pembayaran	Jumlah Bayaran Belum Dibayar	Jumlah Bayaran Dah Dibayar	Tindakan					
A009	Al_Bai	2	0	216.8	0.00	Kemaskini					
← Kembali											

SELECT lp.ID, lp.no_anggota, lp.bilangan_bayaran, lp.jumlah_bayaran

FROM loan_payment lp;

11) Update loan payment

Butiran Pembayaran Pinjaman

Bilangan Pembayaran

Jumlah Bayaran

Kemaskini

Butiran Pembayaran Semasa

Bilangan Pembayaran	0
Jumlah Bayaran Total	0.00

```
UPDATE loan_payment lp JOIN loan_application la ON lp.no_anggota =  
la.no_anggota SET lp.bilangan_bayaran = lp.bilangan_bayaran + 1,  
lp.jumlah_bayaran = lp.jumlah_bayaran + la.bayaran_bulanan WHERE  
lp.no_anggota = :no_anggota;
```

12) List member that wanted to quit

Home Senarai Ahli Senarai Pembiayaan Pemohonan Ahli Pemohonan Pembiayaan

Senarai Ahli

No_anggota	tindakan
A009	More
A009	More

[← Kembali](#)

```
SELECT m.no_anggota, u.name, u.email, u.no_tel, mt.reason1, mt.reason2,  
mt.reason3, mt.reason4, mt.reason5
```



```

FROM Member m JOIN User u ON m.no_anggota = u.no_anggota
JOIN membership_termination mt ON m.no_anggota = mt.no_anggota
WHERE mt.status_member = 'Quit';

```

13) Reason for quitting

Detail Pemohonan Berhenti

Sebab 1	menyesal
Sebab 2	nak guna koperesi terengganu lah
Sebab 3	
Sebab 4	
Sebab 5	

Delete Application

```

SELECT mt.no_anggota, mt.reason1, mt.reason2, mt.reason3, mt.reason4, mt.reason5
FROM membership_termination mt
WHERE mt.status_member = 'Quit';

```

\

ALk Module

1) Pengesahan permohonan

 Home Pengesahan Permohonan Laporan Permohonan							
Permohonan Pembiayaan							
Nama	Email	Jenis Pinjaman	Jumlah Pinjaman	Tempoh Bayaran	Bayaran Bulanan	Remark	Status Permohonan
SYARIFAH DANIA	iaamdsh@gmail.com	Al_Bai	RM 200.00	2 Bulan	RM 9.03	Butiran diisi lengkap	<button>Approve</button> <button>Reject</button>
NAIM DANIEL	imandyyan@gmail.com	Al_Bai	RM 1200.00	2 Bulan	RM 54.20	dokumen terbaik	<button>Approve</button> <button>Reject</button>

[← Kembali](#)

SELECT

loan_application.ID AS loan_app_id,

loan_application.no_anggota,

loan_application.nama_bank_acc,

loan_application.no_bank_acc,

loan_application.jenis_pembiayaan,

loan_application.amaun_dimohon,

loan_application.tempoh_bayaran,

loan_application.jumlah_pembiayaan,

loan_application.bayaran_bulanan,

loan_application.gaji_bersih,

loan_application.gaji_pokok,

loan_status.ID AS loan_status_id,

loan_status.status_loan

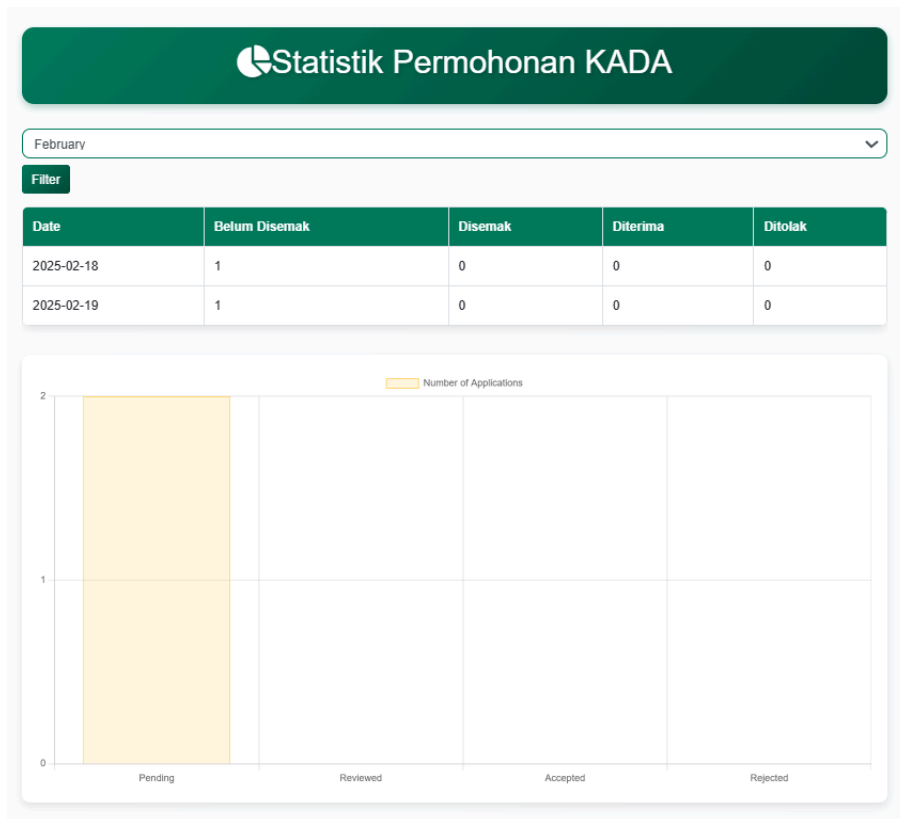
FROM loan_application

LEFT JOIN loan_status ON loan_application.no_anggota = loan_status.no_anggota

WHERE

loan_status.status_loan = '1';

2) Laporan permohonan



```
SELECT * FROM Member
```

```
LEFT JOIN Report ON Member.no_anggota = Report.no_anggota
```

```
WHERE Member.status_member = '1';
```

3) Add staff

Senarai Kerani

Name	Emel	Tindakan
Kada	Kadaclerk@gmail.com	Delete
ALI	mukhriz@graduate.utm.my	Delete
ALI	naim@gmail.com	Delete
dania	dania@gmail.com	Delete

Tambah Kerani

Kembali

+ Tambah Kerani

Nama:

Email:

Kata Laluan:

Please fill out this field.

Hantar

Senarai ALK

Name	Emel	tindakan
kada	kadaAlk@gmail.com	Delete

Tambah ALK

Kembali

Tambah ALK



 Nama

 Email

 Kata Laluan



 Hantar

```
SELECT * FROM ALK;  
  
SELECT * FROM Clerk;  
  
INSERT INTO ALK (ALKID, name, email, password)  
VALUES (  
    '${ALKData["id"]}',  
    '${ALKData["name"]}',  
    '${ALKData["email"]}',  
    '${ALKData["password"]}'  
);
```

```
INSERT INTO Clerk (ClerkID, name, email, password)  
VALUES (  
    '${ClerkData["id"]}',  
    '${ClerkData["name"]}',  
    '${ClerkData["email"]}',  
    '${ClerkData["password"]}'  
);
```


4.2 SQL Statements (DDL & DML)

SQL Statements for User Module

1. UseCase : Register

DDL

```
CREATE TABLE Member ( ID INT AUTO_INCREMENT PRIMARY KEY,
no_anggota VARCHAR(255) UNIQUE NOT NULL, no_PF VARCHAR(255) NOT
NULL, status_member VARCHAR(50) NOT NULL, tarikh DATE NOT NULL );
```

```
CREATE TABLE User ( ID INT AUTO_INCREMENT PRIMARY KEY, no_anggota
VARCHAR(255) NOT NULL, no_PF VARCHAR(255) NOT NULL, name
VARCHAR(255) NOT NULL, email VARCHAR(255) UNIQUE NOT NULL, IC_no
VARCHAR(50) UNIQUE NOT NULL, jantina VARCHAR(10) NOT NULL,
taraf_perkahwinan VARCHAR(50) NOT NULL, agama VARCHAR(50) NOT
NULL, bangsa VARCHAR(50) NOT NULL, jawatan_gred VARCHAR(50) NOT
NULL, no_tel VARCHAR(50) NOT NULL, no_tel_rumah VARCHAR(50),
FOREIGN KEY (no_anggota) REFERENCES Member(no_anggota) ON DELETE
CASCADE );
```

```
CREATE TABLE Address ( ID INT AUTO_INCREMENT PRIMARY KEY,
no_anggota VARCHAR(255) NOT NULL, alamatRumah TEXT NOT NULL,
alamatPejabat TEXT NOT NULL, FOREIGN KEY (no_anggota) REFERENCES
Member(no_anggota) ON DELETE CASCADE );
```

```
CREATE TABLE financial_information ( ID INT AUTO_INCREMENT PRIMARY
KEY, no_anggota VARCHAR(255) NOT NULL, gaji DECIMAL(10,2) NOT NULL,
modal_syer DECIMAL(10,2) NOT NULL, fee_masuk DECIMAL(10,2) NOT
NULL, modal_yuran DECIMAL(10,2) NOT NULL, wang_deposit DECIMAL(10,2)
NOT NULL, tabung_kebajikan DECIMAL(10,2) NOT NULL, simpanan_tetap
DECIMAL(10,2) NOT NULL, FOREIGN KEY (no_anggota) REFERENCES
Member(no_anggota) ON DELETE CASCADE );
```

DML

```
INSERT INTO Member (ID, no_anggota, no_PF, status_member)
VALUES (
'{$MemberData["id"]}',
'{$MemberData["no_anggota"]}',
'{$MemberData["no_PF"]}',
'{$MemberData["status_member"]}'
);
```

```
INSERT INTO User (ID, no_anggota, no_PF, name, email, IC_no, jantina,
taraf_perkahwinan, agama, bangsa, jawatan_gred, no_tel, no_tel_rumah)
```

```
VALUES (
    '{$UserData["id"]}',
    '{$UserData["no_anggota"]}',
    '{$UserData["no_PF"]}',
    '{$UserData["name"]}',
    '{$UserData["email"]}',
    '{$UserData["IC_no"]}',
    '{$UserData["jantina"]}',
    '{$UserData["taraf_perkahwinan"]}',
    '{$UserData["agama"]}',
    '{$UserData["bangsa"]}',
    '{$UserData["jawatan_gred"]}',
    '{$UserData["no_tel"]}',
    '{$UserData["no_tel_rumah"]}'
);
```

```
INSERT INTO financial_information (ID, no_anggota, gaji, modal_syer, fee_masuk,
modal_yuran, wang_deposit, tabung_kebajikan, simpanan_tetap)
```

```
VALUES (
    '{$FinanceData["id"]}',
    '{$FinanceData["no_anggota"]}',
    '{$FinanceData["gaji"]}',
    '{$FinanceData["modal_syer"]}',
    '{$FinanceData["fee_masuk"]}',
    '{$FinanceData["modal_yuran"]}',
    '{$FinanceData["wang_deposit"]}',
    '{$FinanceData["tabung_kebajikan"]}',
    '{$FinanceData["simpanan_tetap"]}'
);
```

```
INSERT INTO Address (ID, no_anggota, alamatRumah, alamatPejabat)
```

```
VALUES (
    '{$AddressData["id"]}',
    '{$AddressData["no_anggota"]}',
```



```
'{$AddressData["alamatRumah"]}',
'{$AddressData["alamatPejabat"]}'
);
```

2. UseCase : Manage Profile

DML

SELECT

```
-- User table fields
u.name AS "Nama",
u.IC_no AS "Kad Pengenalan",
u.jantina AS "Jantina",
u.agama AS "Agama",
u.bangsa AS "Bangsa",
u.taraf_perkahwinan AS "Taraf Perkahwinan",
u.gaji AS "Gaji",
u.email AS "Emel",
u.no_tel AS "No Tel Bimbit",
u.no_tel_rumah AS "No Tel Rumah",
```

```
-- Member table fields
m.no_anggota AS "No Anggota",
m.no_PF AS "No PF",
u.jawatan_gred AS "Jawatan & Gred",
u.alamat_pejabat AS "Alamat Pejabat",
u.alamat_rumah AS "Alamat Rumah"
```

FROM

```
User u
```

JOIN

```
Member m ON u.no_anggota = m.no_anggota
```

WHERE

```
u.IC_no = '{IC_no}' OR m.no_anggota = '{no_anggota}' OR u.email = '{email}';
```

SQL Statements for loan Application Module

1. UseCase : applying a loan

DDL

```
CREATE TABLE loan_application ( ID INT AUTO_INCREMENT PRIMARY KEY,
no_anggota VARCHAR(255) NOT NULL, nama_bank_acc VARCHAR(255) NOT
NULL, no_bank_acc VARCHAR(50) NOT NULL, jenis_pembiayaan
VARCHAR(50) NOT NULL, amaun_dimohon DECIMAL(10,2) NOT NULL,
tempoh_bayaran INT NOT NULL, jumlah_pembiayaan DECIMAL(10,2) NOT
NULL, bayaran_bulanan DECIMAL(10,2) NOT NULL, gaji_bersih DECIMAL(10,2)
NOT NULL, gaji_pokok DECIMAL(10,2) NOT NULL, FOREIGN KEY
(no_anggota) REFERENCES Member(no_anggota) ON DELETE CASCADE );
```

```
CREATE TABLE loan_status ( ID INT AUTO_INCREMENT PRIMARY KEY,
no_anggota VARCHAR(255) NOT NULL, status_loan VARCHAR(50) NOT NULL,
FOREIGN KEY (no_anggota) REFERENCES Member(no_anggota) ON DELETE
CASCADE );
```

```
CREATE TABLE Penjamin ( ID INT AUTO_INCREMENT PRIMARY KEY,
no_anggota VARCHAR(255) NOT NULL, guarantor_name VARCHAR(255) NOT
NULL, No_KP VARCHAR(50) NOT NULL, no_anggota_guarantor
VARCHAR(255) NOT NULL, no_PF VARCHAR(255) NOT NULL, FOREIGN
KEY (no_anggota) REFERENCES Member(no_anggota) ON DELETE CASCADE
);
```

DML

```
INSERT INTO loan_application (ID, no_anggota, nama_bank_acc, no_bank_acc,
jenis_pembiayaan, amaun_dimohon, tempoh_bayaran, jumlah_pembiayaan,
bayaran_bulanan, gaji_bersih, gaji_pokok)
```

```
VALUES      ('{$LoanData["id"]}',          '{$LoanData["no_anggota"]}',
'{$LoanData["nama_bank_acc"]}',            '{$LoanData["no_bank_acc"]}',
'{$LoanData["jenis_pembiayaan"]}',          '{$LoanData["amaun_dimohon"]}',
'{$LoanData["tempoh_bayaran"]}',            '{$LoanData["jumlah_pembiayaan"]}',
'{$LoanData["bayaran_bulanan"]}',           '{$LoanData["gaji_bersih"]}',
'{$LoanData["gaji_pokok"]}');
```

```
INSERT INTO loan_status (ID, no_anggota, status_loan)
```

```
VALUES      ('{$LoanStatusData["id"]}',      '{$LoanStatusData["no_anggota"]}',
'{$LoanStatusData["status_loan"]}');
```

```
INSERT INTO loan_payment (ID, no_anggota, bilangan_bayaran, jumlah_bayaran)
```

```
VALUES ('{$LoanPaymentData["id"]}', '{$LoanPaymentData["no_anggota"]}',
'{$LoanPaymentData["bilangan_bayaran"]}',
'{$LoanPaymentData["jumlah_bayaran"]}');
```

2. UseCase : Check Loan Status

DML

SELECT

m.m_app_id,

m.m_user_id,

u.IC_no,

u.name,

m.m_status

FROM

membership m

JOIN

user u ON m.m_user_id = u.ID;

SELECT

sp.modal_syer_total AS "Modal Syer",

sp.modal_yuran_total AS "Modal Yuran",

sp.simpanan_anggota_total AS "Simpanan Anggota",

sp.tabung_kebajikan_total AS "Tabung Anggota",

sp.wang_deposit_total AS "Simpanan Tetap",

lp1.jumlah_bayaran AS "Al-Bai",

lp2.jumlah_bayaran AS "Al-Innah",

lp3.jumlah_bayaran AS "Pulih Kenderaan",

lp4.jumlah_bayaran AS "Khas",

lp5.jumlah_bayaran AS "Al-Qadrul Hassan"

FROM

Saving_Payment sp

LEFT JOIN loan_payment lp1 ON sp.no_anggota = lp1.no_anggota AND lp1.loan_type = 'Al-Bai'

LEFT JOIN loan_payment lp2 ON sp.no_anggota = lp2.no_anggota AND lp2.loan_type = 'Al-Innah'

LEFT JOIN loan_payment lp3 ON sp.no_anggota = lp3.no_anggota AND lp3.loan_type = 'Pulih Kendaraan'

LEFT JOIN loan_payment lp4 ON sp.no_anggota = lp4.no_anggota AND lp4.loan_type = 'Khas'

LEFT JOIN loan_payment lp5 ON sp.no_anggota = lp5.no_anggota AND lp5.loan_type = 'Al-Qadrul Hassan'

WHERE

sp.no_anggota = '{no_anggota}';

SQL Statements for Clerk Module

1. UseCase : Create Account

DML

SELECT

m.m_app_id,

m.m_user_id,

u.IC_no,

u.name,

m.m_status

FROM

membership m

JOIN

user u ON m.m_user_id = u.ID;

UPDATE membership SET m_status = 'Accepted' WHERE id = :id;

UPDATE membership SET m_status = 'Rejected' WHERE m_app_id = :id;

2. UseCase : Manage list of Member

DDL

```
CREATE TABLE membership_termination ( ID INT AUTO_INCREMENT  
PRIMARY KEY, no_anggota VARCHAR(255) NOT NULL, reason TEXT NOT  
NULL, FOREIGN KEY (no_anggota) REFERENCES Member(no_anggota) ON  
DELETE CASCADE );
```

DML

```
SELECT m.no_anggota, u.name, u.email, u.no_tel, u.jawatan_gred
```

```
FROM Member m
```

```
JOIN User u ON m.no_anggota = u.no_anggota
```

```
JOIN membership_termination mt ON m.no_anggota = mt.no_anggota
```

```
WHERE mt.status_member != 'Quit';
```

```
SELECT m.no_anggota, m.name, u.no_IC, u.jantina, u. agama, u.bangsa, u.  
taraf_perkahwinan, u.alamat_rumah, u.poskod, u.negeri, u.no_PF, u.email, u.no_tel,  
u.jawatan_gred, fi.gaji, fi.modal_syer, fi.fee_masuk, fi.modal_yuran, fi.wang_deposit,  
fi.tabung_kebajikan, fi.simpanan_tetap
```

```
FROM Member m
```

```
JOIN User u ON m.no_anggota = u.no_anggota
```

```
JOIN financial_information fi ON m.no_anggota = fi.no_anggota;
```

```

SELECT  m.no_anggota, u.name, u.email, u.no_tel, mt.reason1, mt.reason2,
mt.reason3, mt.reason4, mt.reason5

FROM Member m JOIN User u ON m.no_anggota = u.no_anggota

JOIN membership_termination mt ON m.no_anggota = mt.no_anggota

WHERE mt.status_member = 'Quit';

```

3. UseCase : Manage Loan Application

DML

```

SELECT  la.ID,      la.no_anggota,      la.nama_bank_acc,      la.no_bank_acc,
la.jenis_pembiayaan, la.amaun_dimohon, la.tempoh_bayaran, la.jumlah_pembiayaan

FROM loan_application la

JOIN loan_status ls ON la.ID = ls.ID

WHERE ls.status_loan = 'Pending';

```

```

DELETE FROM loan_application

```

```

WHERE ID IN (

```

```

    SELECT la.ID

```

```

    FROM loan_application la

```

```

    JOIN loan_payment lp ON la.no_anggota = lp.no_anggota

```

```

    WHERE lp.bilangan_bayaran = lp.jumlah_bayaran -- or any other condition that
indicates full payment

```

```

);

```

4. UseCase : Interest

DDL

```
CREATE TABLE faedah ( ID INT AUTO_INCREMENT PRIMARY KEY, peratus  
DECIMAL(5,2) NOT NULL );
```

DML

```
SELECT
```

```
    f.ID AS "ID",
```

```
    f.peratus AS "Peratus"
```

```
FROM
```

```
    faedah f
```

```
WHERE
```

```
    f.ID = '{ID}';
```

```
UPDATE faedah
```

```
SET peratus = '{peratus}'
```

```
WHERE ID = '{ID}';
```

5. UseCase : Payment

DDL

```
CREATE TABLE loan_payment ( ID INT AUTO_INCREMENT PRIMARY KEY,  
no_anggota VARCHAR(255) NOT NULL, bilangan_bayaran INT NOT NULL,  
jumlah_bayaran DECIMAL(10,2) NOT NULL, FOREIGN KEY (no_anggota)  
REFERENCES Member(no_anggota) ON DELETE CASCADE );
```

```
CREATE TABLE Saving_Payment ( ID INT AUTO_INCREMENT PRIMARY KEY,  
no_anggota VARCHAR(255) NOT NULL, modal_syer_total DECIMAL(10,2) NOT  
NULL, tabung_kebajikan_total DECIMAL(10,2) NOT NULL,  
simpanan_anggota_total DECIMAL(10,2) NOT NULL, wang_deposit_total  
DECIMAL(10,2) NOT NULL, modal_yuran_total DECIMAL(10,2) NOT NULL,  
FOREIGN KEY (no_anggota) REFERENCES Member(no_anggota) ON DELETE  
CASCADE );
```

DML

```
SELECT
```

```

    u.name,
    u.email,
    sp.no_anggota,
    sp.modal_syer_total,
    sp.tabung_kebajikan_total,
    sp.simpanan_anggota_total,
    sp.wang_deposit_total,
    sp.modal_yuran_total
FROM
    Saving_Payment sp
JOIN
    Member m ON sp.no_anggota = m.no_anggota
JOIN
    User u ON m.no_anggota = u.no_anggota;

SELECT lp.ID, lp.no_anggota, lp.bilangan_bayaran, lp.jumlah_bayaran
FROM loan_payment lp;

UPDATE Saving_Payment
SET modal_syer_total = {modal_syer_total},
    tabung_kebajikan_total = {tabung_kebajikan_total},
    simpanan_anggota_total = {simpanan_anggota_total},
    wang_deposit_total = {wang_deposit_total},
    modal_yuran_total = {modal_yuran_total}
WHERE no_anggota = no_anggota;

```



```
UPDATE loan_payment lp JOIN loan_application la ON lp.no_anggota =  
la.no_anggota SET lp.bilangan_bayaran = lp.bilangan_bayaran + 1,  
lp.jumlah_bayaran = lp.jumlah_bayaran + la.bayaran_bulanan WHERE  
lp.no_anggota = :no_anggota;
```

SQL Statements for ALK Module

1. UseCase : Verify loan application

DDL

```
CREATE TABLE loan_application (  
    ID INT AUTO_INCREMENT PRIMARY KEY,  
    no_anggota VARCHAR(255) NOT NULL,  
    nama_bank_acc VARCHAR(255) NOT NULL,  
    no_bank_acc VARCHAR(255) NOT NULL,  
    jenis_pembiayaan VARCHAR(255) NOT NULL,  
    amaun_dimohon DECIMAL(10,2) NOT NULL,  
    tempoh_bayaran INT NOT NULL,  
    jumlah_pembiayaan DECIMAL(10,2) NOT NULL,  
    bayaran_bulanan DECIMAL(10,2) NOT NULL,  
    gaji_bersih DECIMAL(10,2) NOT NULL,  
    gaji_pokok DECIMAL(10,2) NOT NULL,  
    FOREIGN KEY (no_anggota) REFERENCES Member(no_anggota) ON DELETE  
    CASCADE  
);
```

DML

```
SELECT  
    loan_application.ID AS loan_app_id,  
    loan_application.no_anggota,  
    loan_application.nama_bank_acc,  
    loan_application.no_bank_acc,  
    loan_application.jenis_pembiayaan,  
    loan_application.amaun_dimohon,  
    loan_application.tempoh_bayaran,
```

```

loan_application.jumlah_pembiayaan,
loan_application.bayaran_bulanan,
loan_application.gaji_bersih,
loan_application.gaji_pokok,
loan_status.ID AS loan_status_id,
loan_status.status_loan
FROM loan_application
LEFT JOIN loan_status ON loan_application.no_anggota = loan_status.no_anggota
WHERE loan_status.status_loan = '1';

```

2. UseCase : View annual report

DDL

```

CREATE TABLE Report ( ID INT AUTO_INCREMENT PRIMARY KEY,
no_anggota VARCHAR(255) NOT NULL, status_member VARCHAR(50) NOT
NULL, date DATE NOT NULL, FOREIGN KEY (no_anggota) REFERENCES
Member(no_anggota) ON DELETE CASCADE );

```

DML

```

SELECT * FROM Member
LEFT JOIN Report ON Member.no_anggota = Report.no_anggota
WHERE Member.status_member = '1';

```

3. UseCase : Add Staff

DDL

```
CREATE TABLE ALK ( ALKID INT AUTO_INCREMENT PRIMARY KEY, name
VARCHAR(255) NOT NULL, email VARCHAR(255) UNIQUE NOT NULL,
password VARCHAR(255) NOT NULL );
```

```
CREATE TABLE Clerk ( ClerkID INT AUTO_INCREMENT PRIMARY KEY, name
VARCHAR(255) NOT NULL, email VARCHAR(255) UNIQUE NOT NULL,
password VARCHAR(255) NOT NULL );
```

DML

```
SELECT * FROM ALK;
```

```
SELECT * FROM Clerk;
```

```
INSERT INTO ALK (ALKID, name, email, password)
```

```
VALUES (
```

```
'{$ALKData["id"]}',
```

```
'{$ALKData["name"]}',
```

```
'{$ALKData["email"]}',
```

```
'{$ALKData["password"]}'
```

```
);
```

```
INSERT INTO Clerk (ClerkID, name, email, password)
```

```
VALUES (
```

```
'{$ClerkData["id"]}',
```

```
'{$ClerkData["name"]}',
```

```
'{$ClerkData["email"]}',
```

```
'{$ClerkData["password"]}'
```

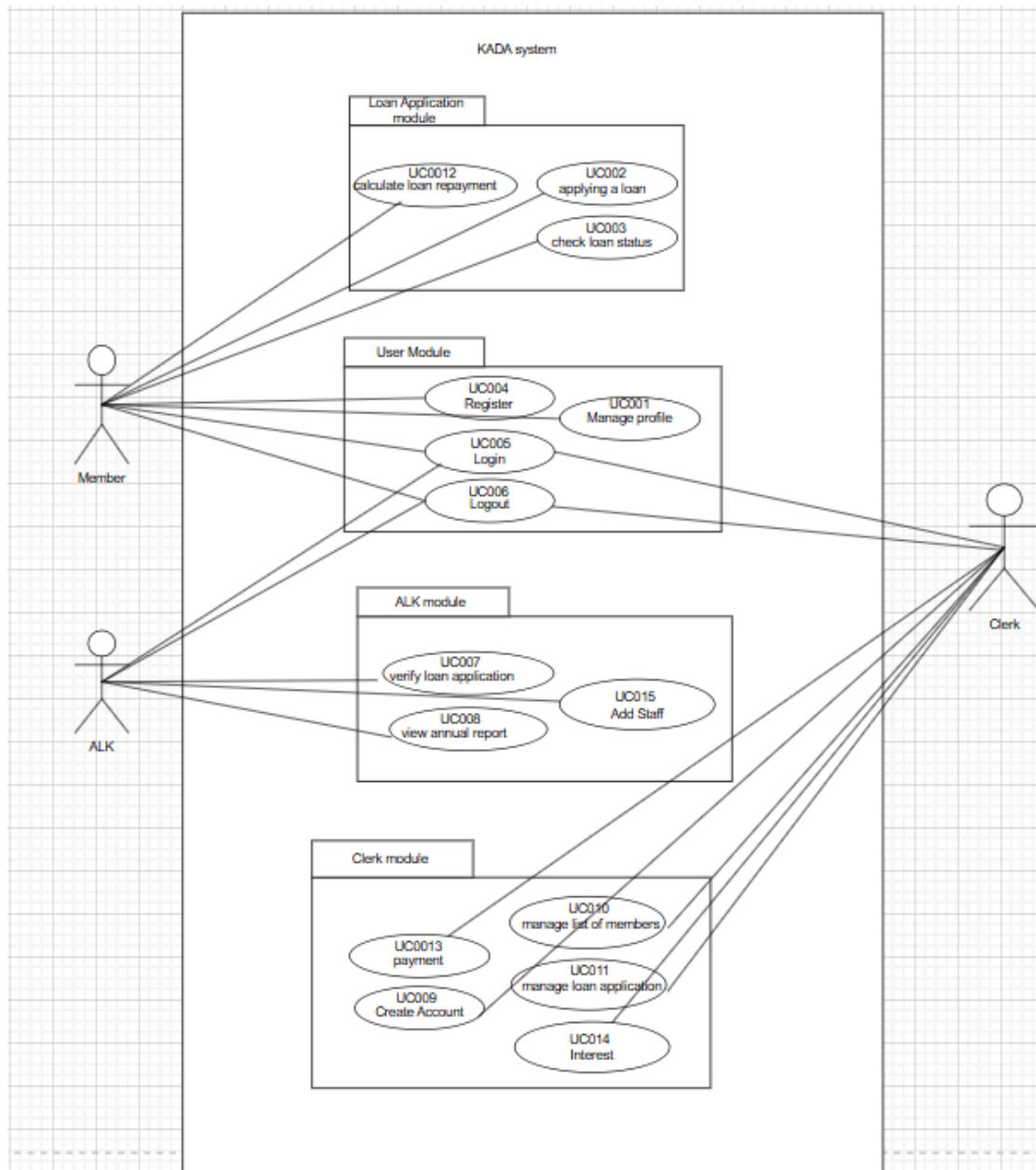
```
);
```


Summary

In conclusion, this WBL effort successfully solves the issues that the Kemubu Agricultural Development Board (KADA) cooperative had previously faced. These challenges were carefully examined, and a web-based system was created to offer useful solutions. The system was constructed using a variety of technologies, guaranteeing increased effectiveness and functionality. Additionally, a methodical strategy was used to develop a well-structured database. A conceptual design that defined entities, relationships, and qualities was the first step in the process. The next step was logical database design, which involved identifying foreign keys to create distinct relationships between data. Finally, normalization was used to preserve data consistency, remove redundancy, and improve database flexibility. Following completion, the database was integrated with the created web application and put into use in a database management system.

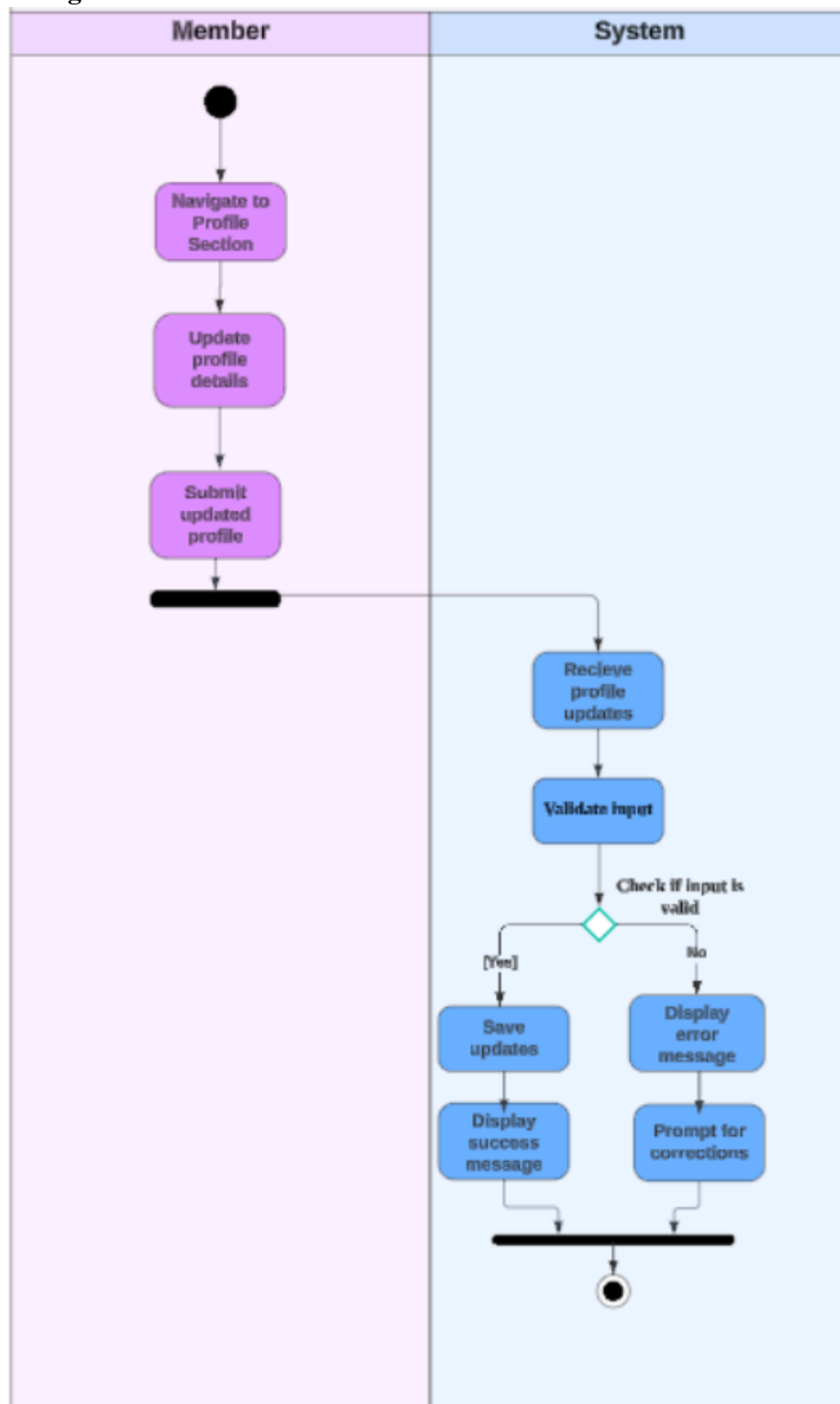
Appendices

6.1 Use Case Diagram for the system.

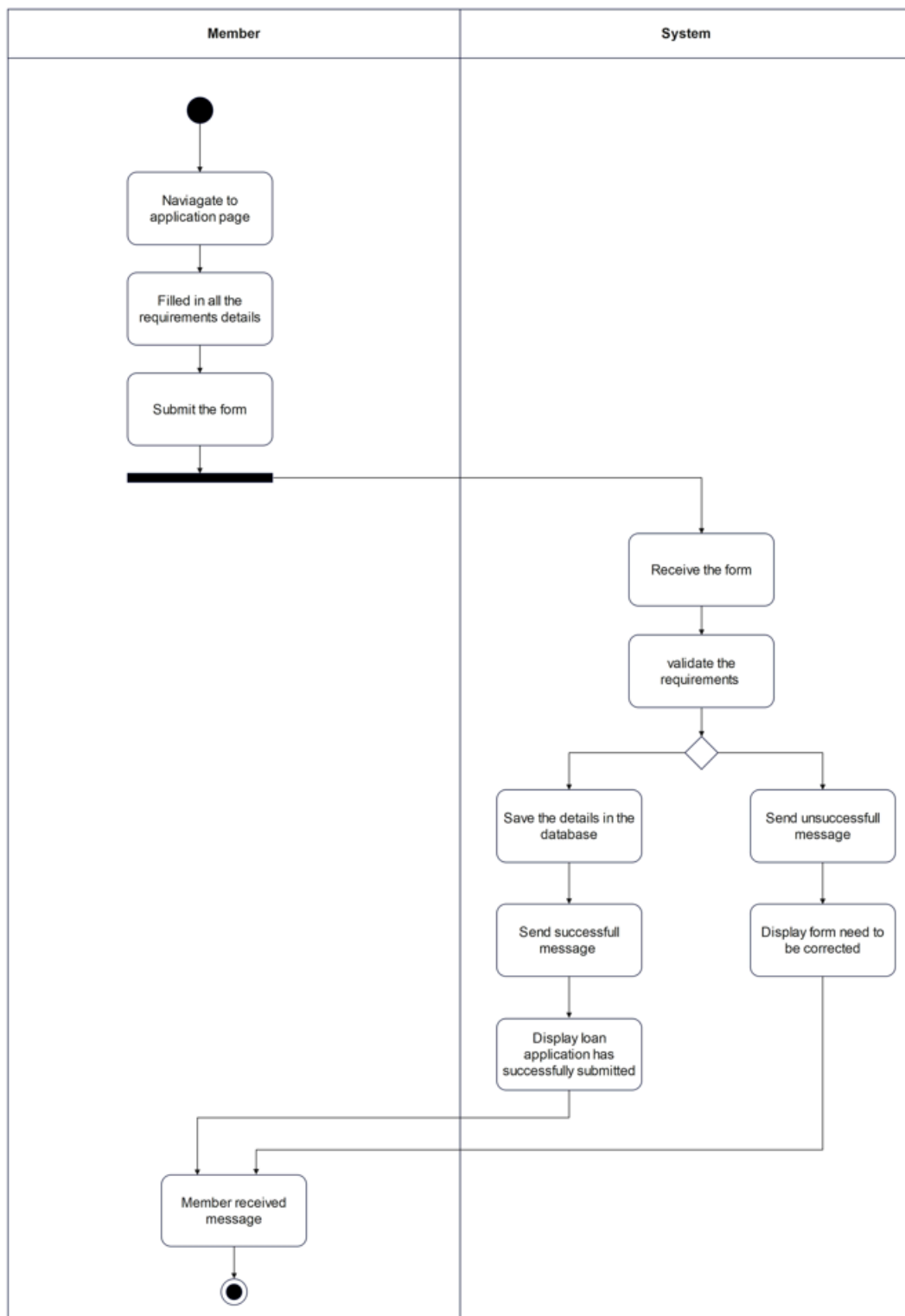


6.2 Activity Diagrams for every use case identified.

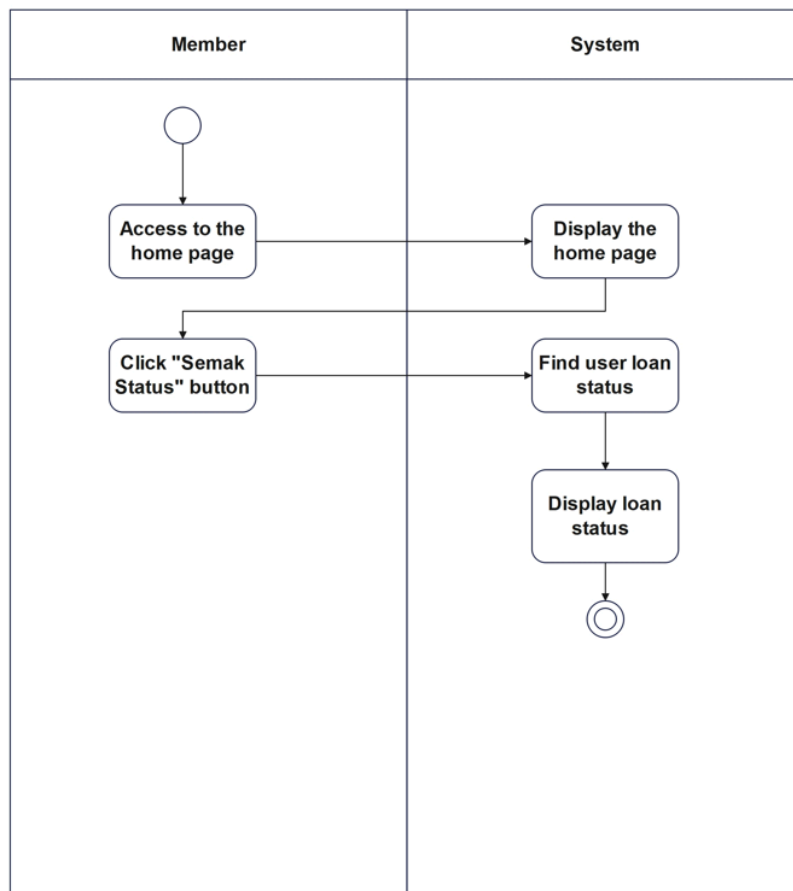
UC001 <Manage Profile>



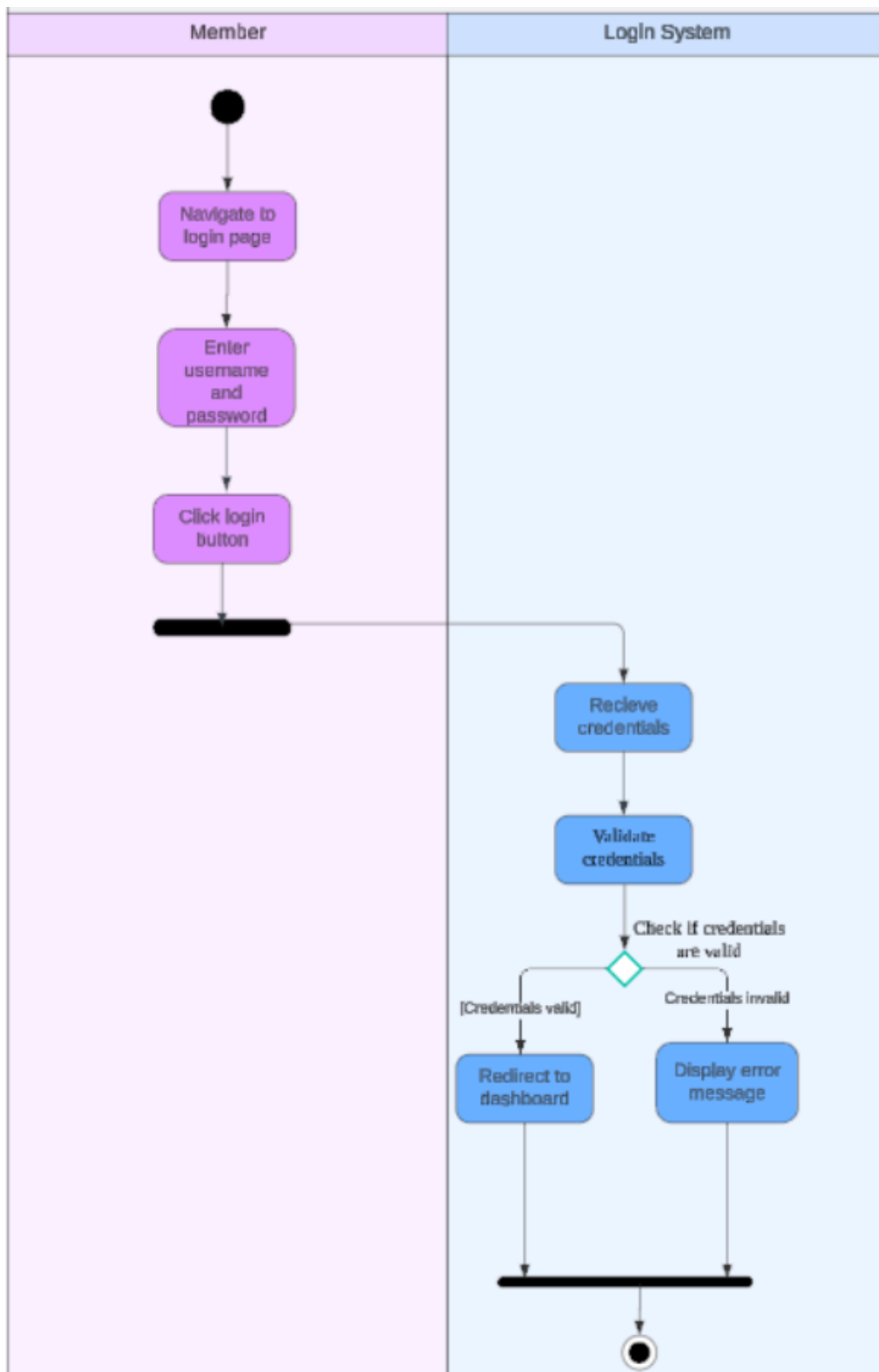
UC002 <Applying Loan>



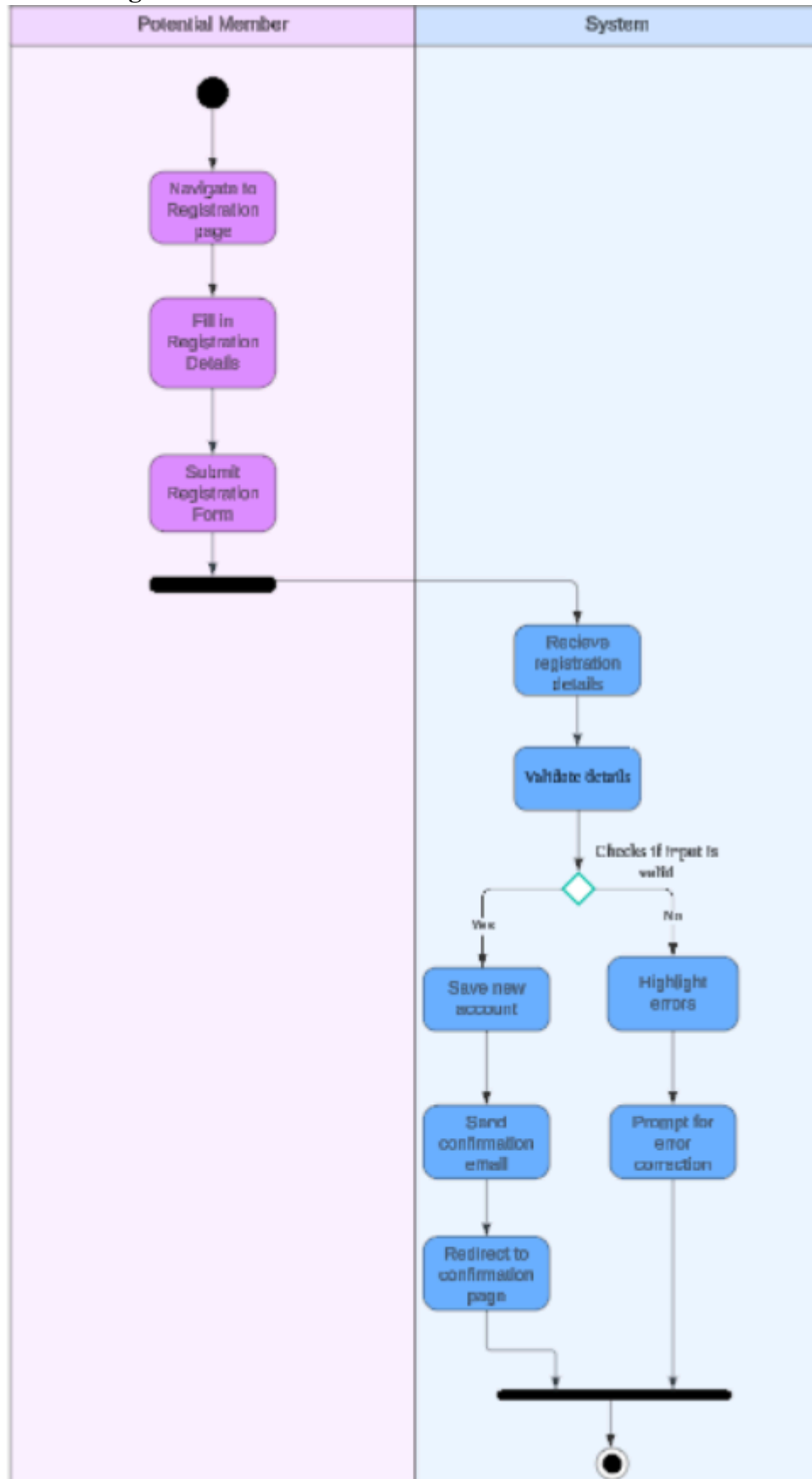
UC003 <Check Loan Status>



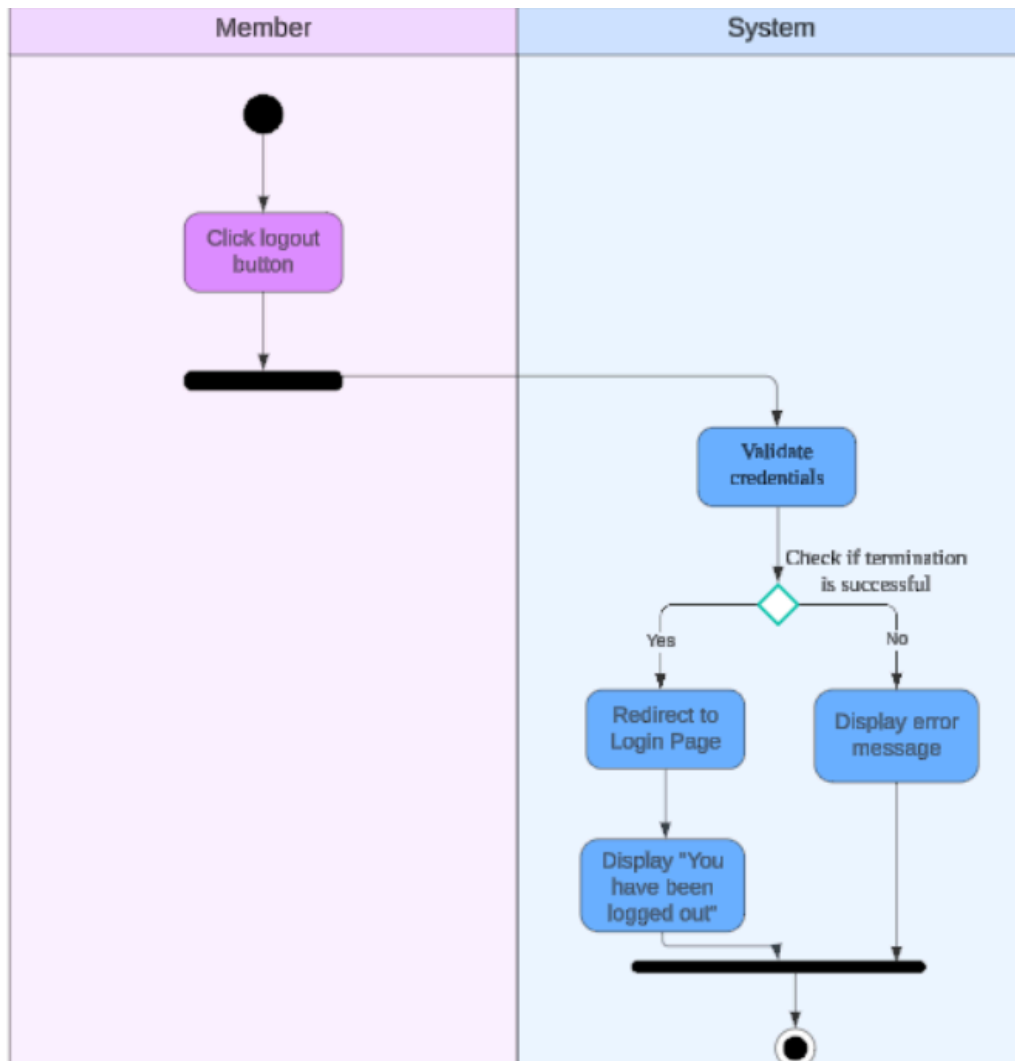
UC004 <Register>



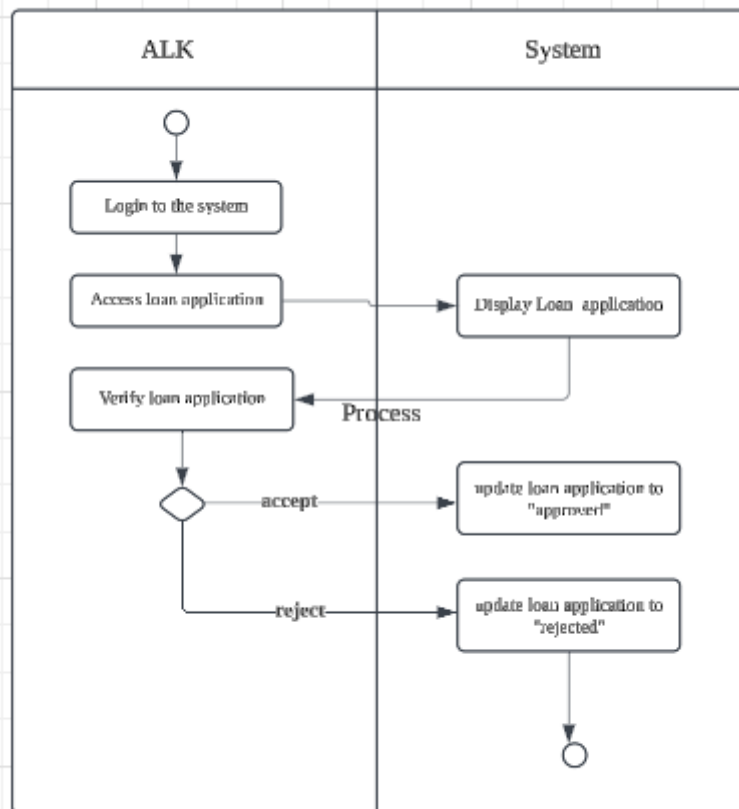
UC005 <Login>



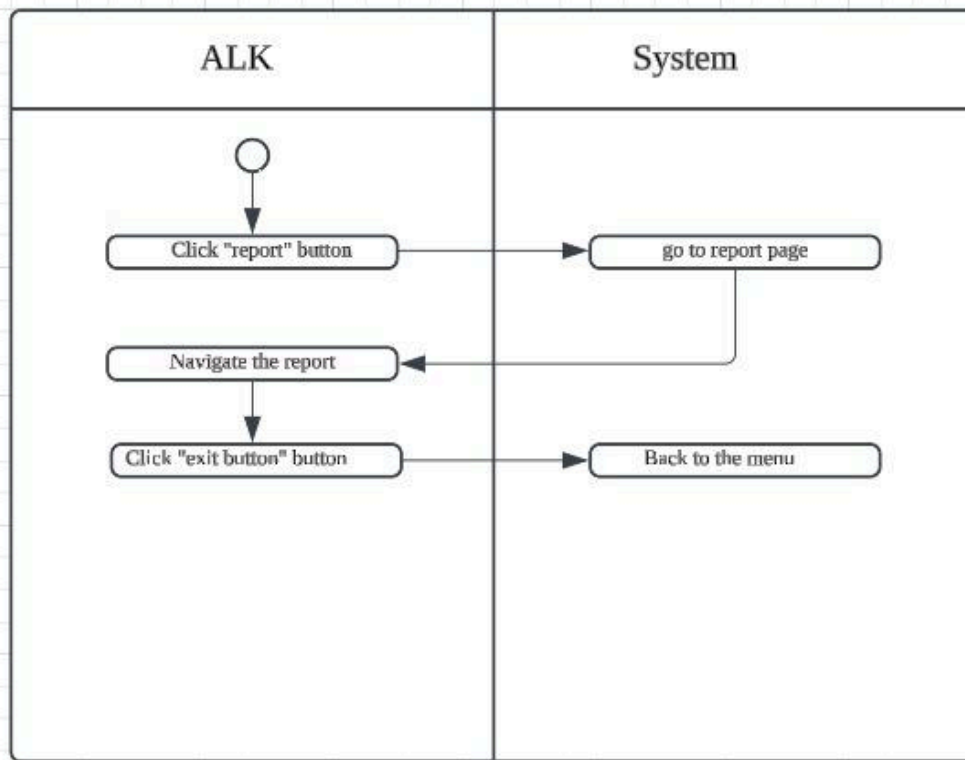
UC006 <Logout>



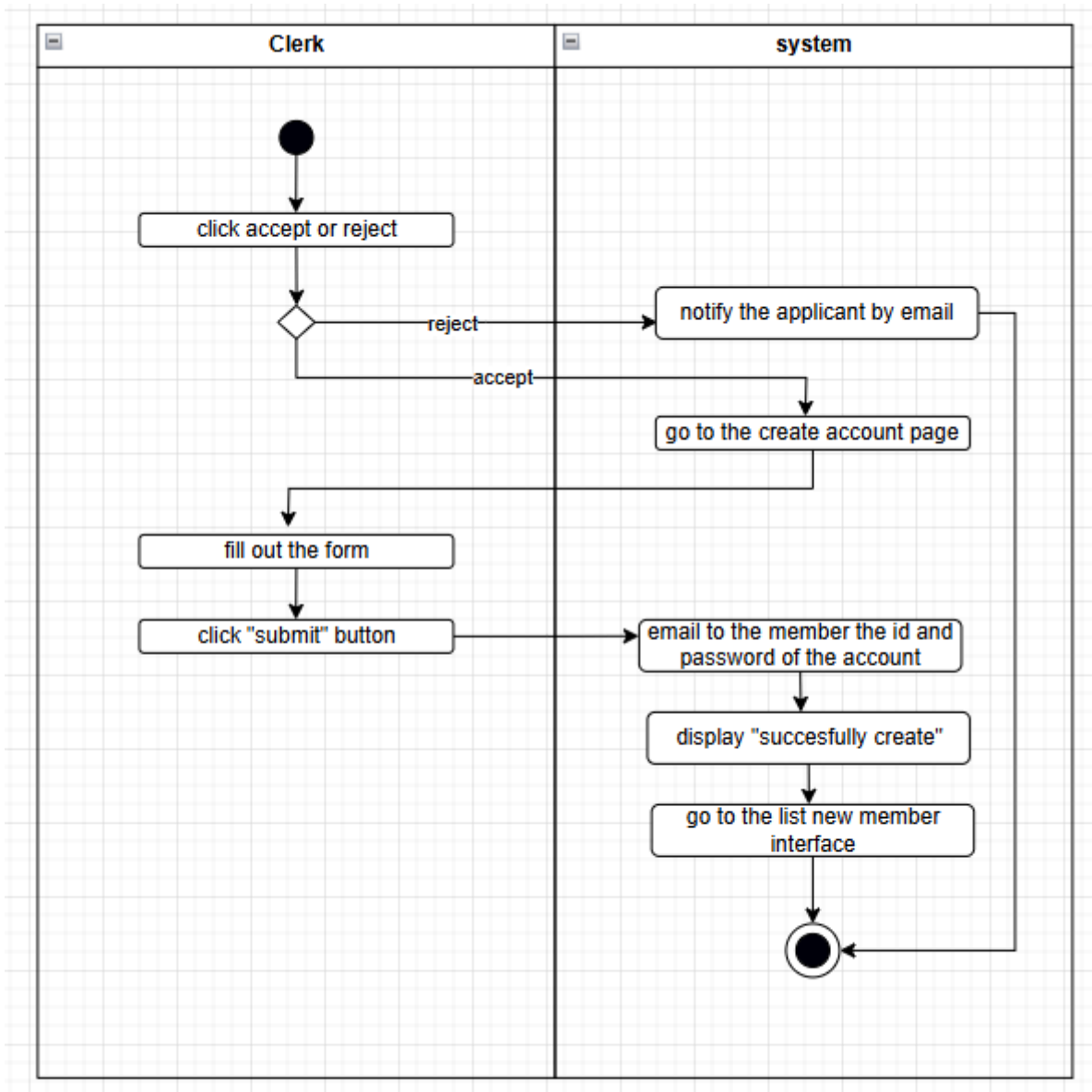
UC007 <Verify Loan Application>



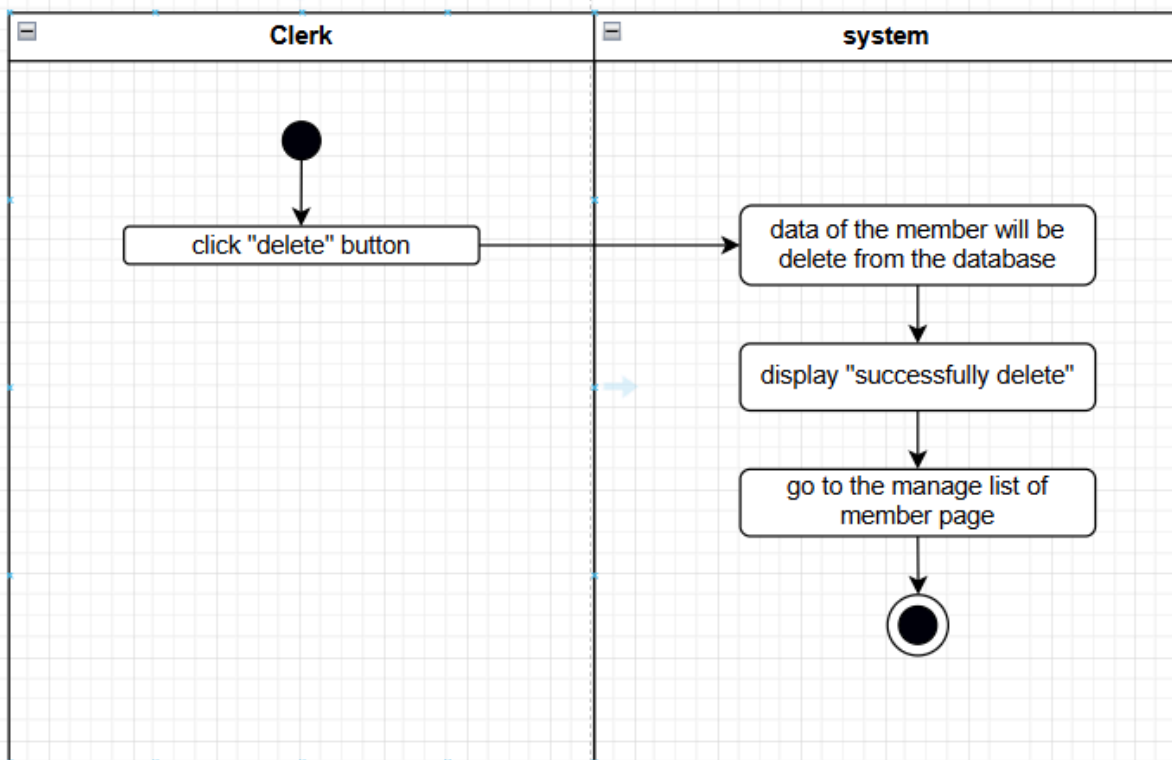
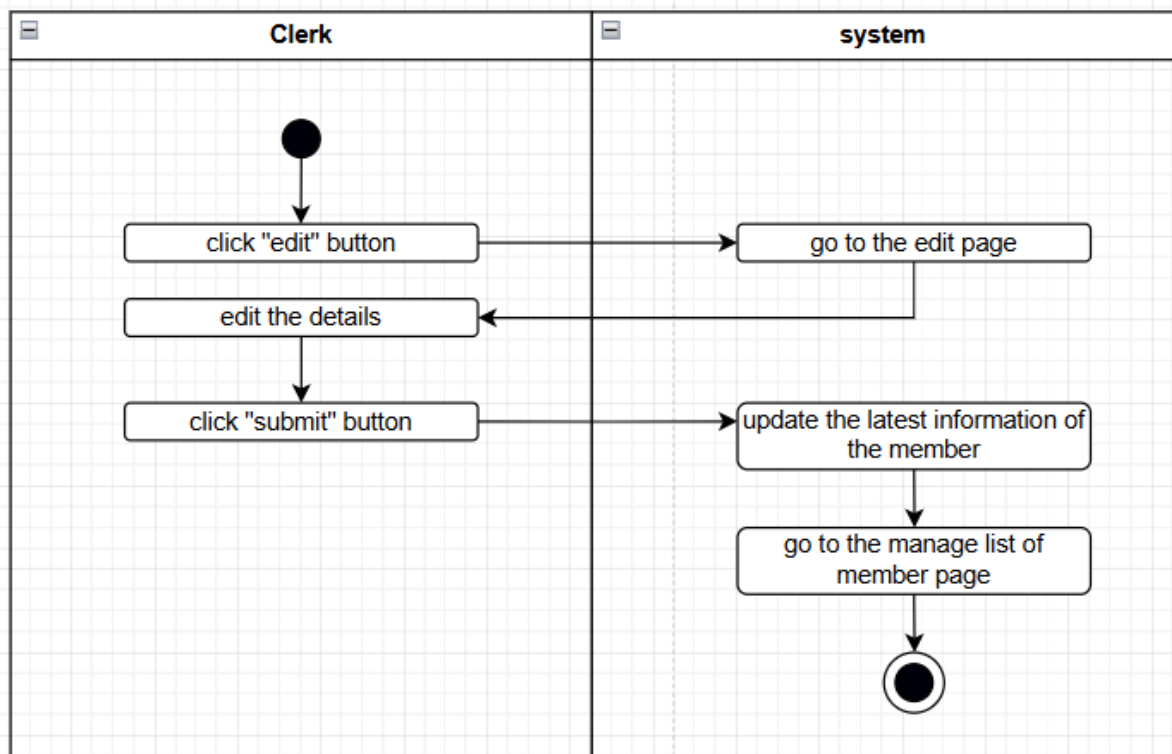
UC008 <View Annual Report>

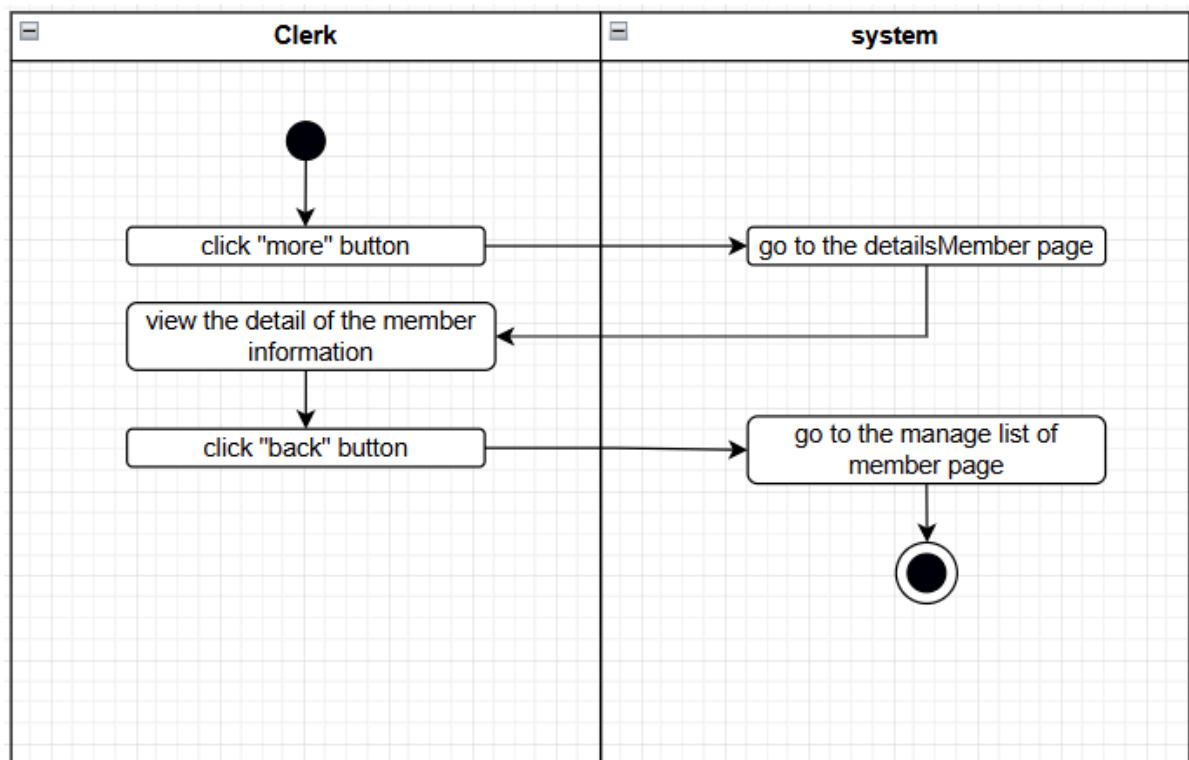


UC009 <Create an Account>

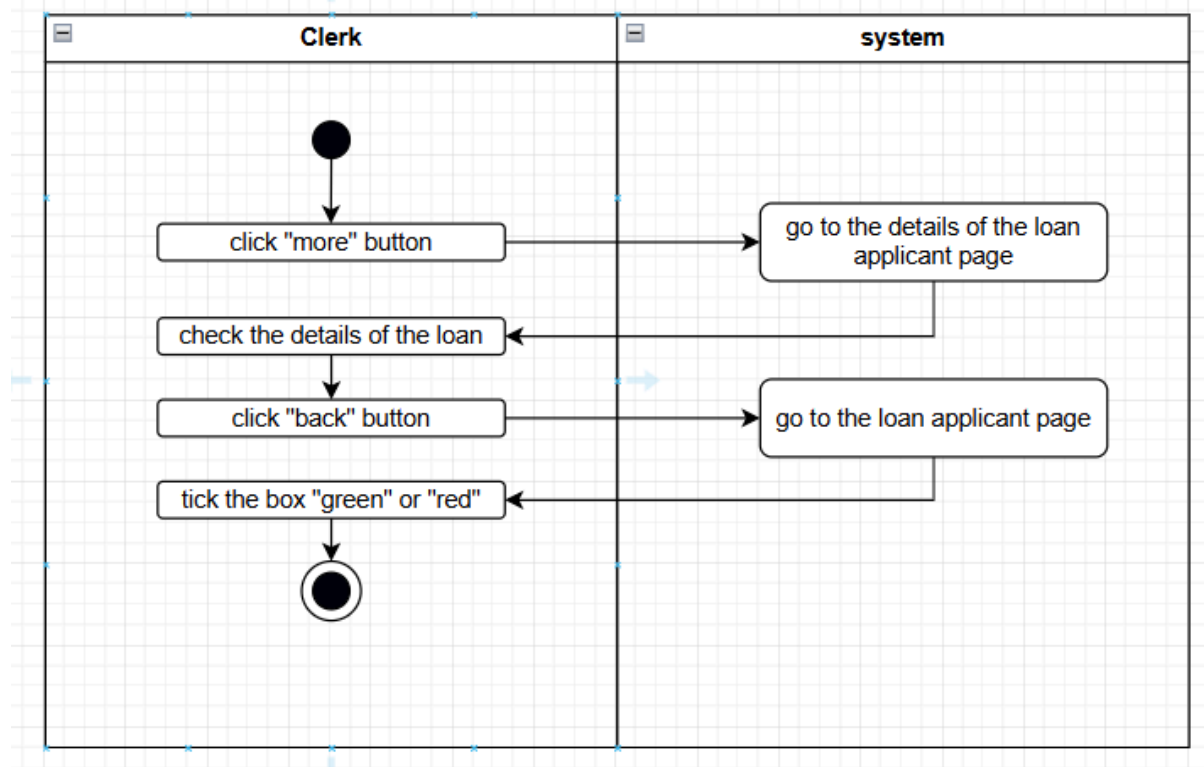


UC010 <Manage List of Members>

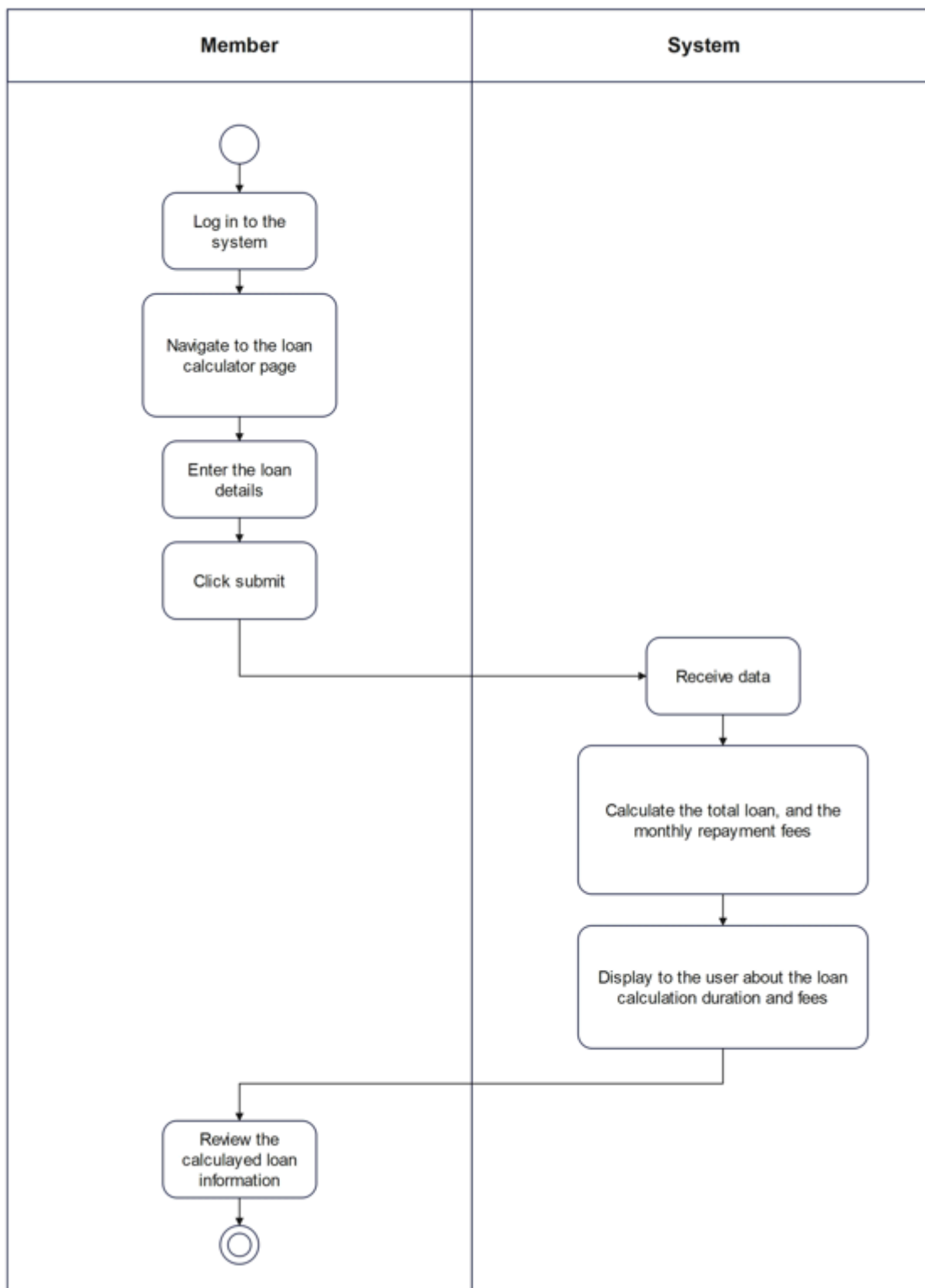




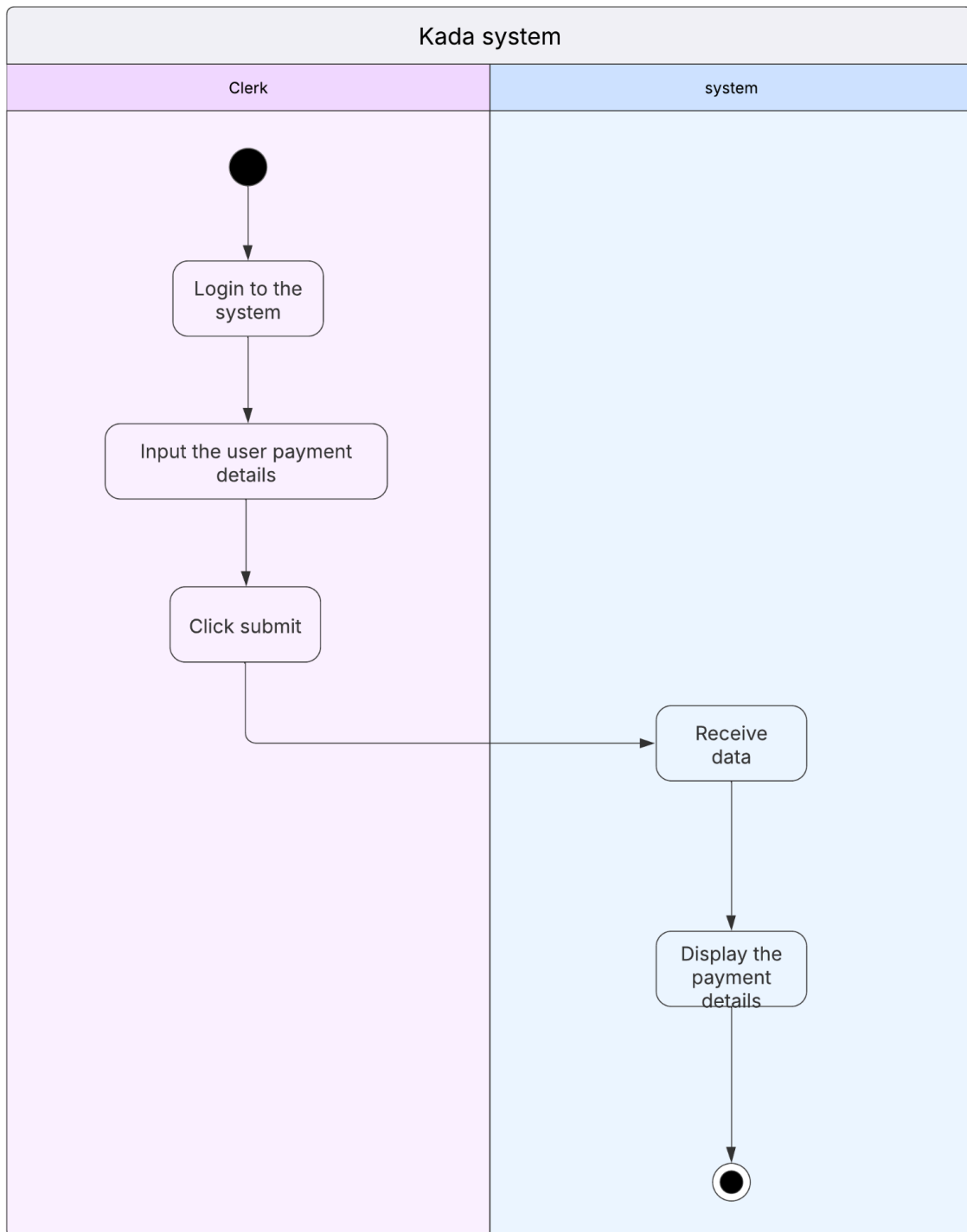
UC011 <Manage Loan Application>



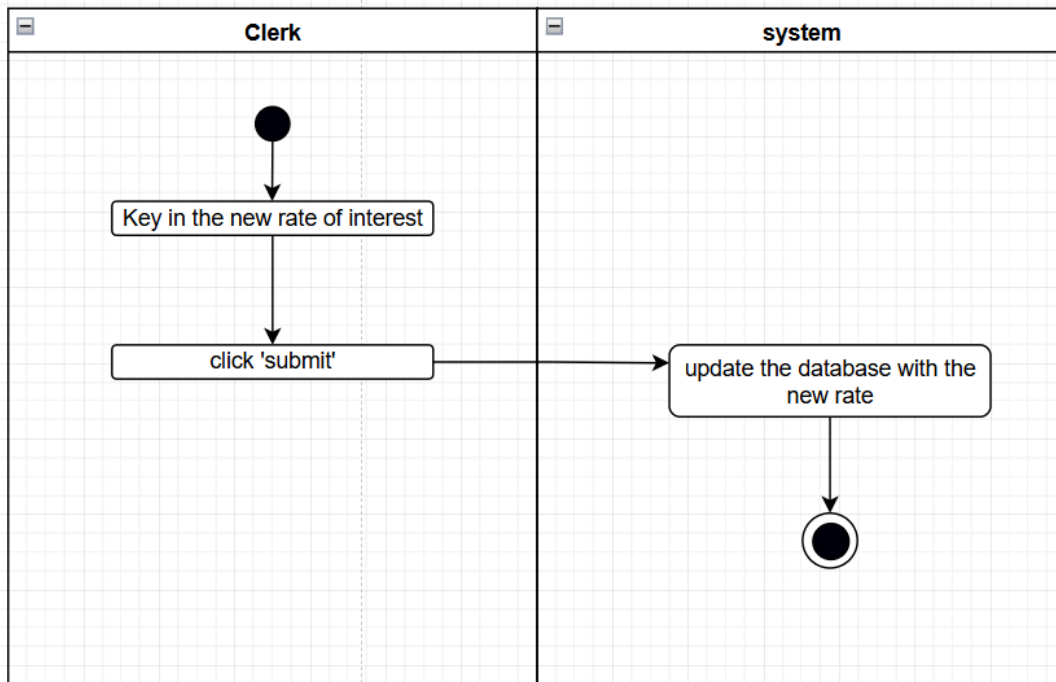
UC012 <Calculate loan Payment>



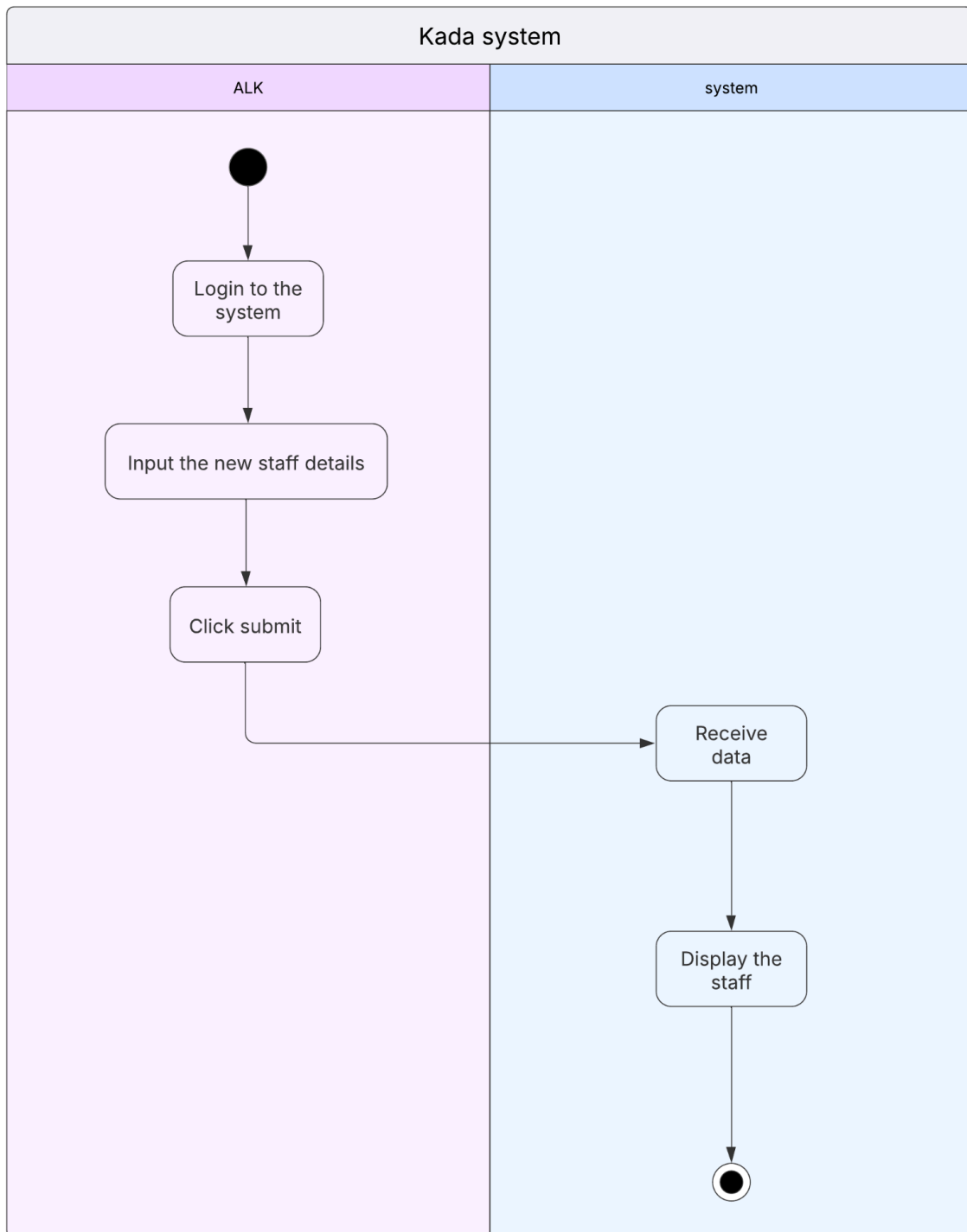
UC013 <Payment>



UC014 <Interest>



UC015 <Add Staff>



6.3 Meeting Log

DATE/TIME	17/2/2024		
LOCATION	ONLINE MEETING		
AGENDA	Completing the P3 logical design report and finalizing all information.		
Meeting MC	SYARIFAH DANIA BINTI SYED ABU BAKAR		
ATTENDANCE			
NAME	TIME	REASON FOR ABSENCE	
MUHAMMAD NAIM BIN ABDULLAH	2:00 PM	-	
MUHAMMAD MUKHRITZ AL IMAN BIN MOHD RAFFI	2:00 PM	-	
SYARIFAH DANIA BINTI SYED ABU BAKAR ;	2:00 PM	-	
YASMIN BATRISYIA BIN ZAHIRUDDIN	2:00 PM	-	
MINUTES			
NO.	ITEM DISCUSSED	IDEA/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	P3 Logical Design Report Finalization	Discussed completing all pending sections of the logical design report. Suggestion to divide the sections among the team members for faster completion. (SYARIFAH DANIA BINTI SYED ABU BAKAR)	SYARIFAH DANIA BINTI SYED ABU BAKAR to ensure report sections are finalized by 20/2/2024.

2	Review of Information Accuracy	Suggested reviewing the data for accuracy and consistency across sections before finalizing. (MUHAMMAD NAIM BIN ABDULLAH)	MUHAMMAD NAIM BIN ABDULLAH to cross-check all data entries by 19/2/2024.
3	Pending Tasks	Discussed remaining tasks and who will be responsible for each. (YASMIN BATRISYIA BIN ZAHIRUDDIN)	YASMIN BATRISYIA BIN ZAHIRUDDIN to update task list and ensure completion by 20/2/2024.
4	Additional Review of Logical Design	Muhammad Naim Bin Abdullah suggested reviewing the logical design for potential inconsistencies and ensuring alignment across all sections. (MUHAMMAD NAIM BIN ABDULLAH)	MUHAMMAD NAIM BIN ABDULLAH to review logical design for consistency by 19/2/2024.